# SOME FUNDAMENTAL GEOMETRICAL PROPERTIES OF PLANE SETS OF FRACTIONAL DIMENSIONS

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## 1. Introduction

1.1. NOTATION AND DEFINITIONS. Given any positive number $q$, by $A(q)$ we denote any sequence of convex areas $U$, each of diameter $dU < q$.

Suppose that we have a plane set of points $E$, and that $A(E,q)$ denotes any set $A(q)$ which contains $E$. Then by

$$\Lambda_q^s E$$

we denote the lower bound of

$$\sum_{A(E,q)} (dU)^s$$

taken over all possible sets $A(E,q)$.

The *outer Hausdorff s-dimensional measure* of $E$ is given by

$$\Lambda^s E = \lim_{q \to 0} \Lambda_q^s E.$$

It can be shown that $\Lambda^s E$ is an outer measure in the sense of Carathéodory (Saks (**14**), § 8, 53). A set $E$ is *s-measurable* if it is measurable with respect to this measure. It is an *s-set* if in addition $\Lambda^s E$ is positive and finite. Any $s$-set, where $s = 1$, is a *linearly measurable set.*

Where there is no ambiguity the term *s-dimensional measure* will be replaced by *measure.* Also, *almost all* points of any given set will mean all points of the set with the exception of a set of measure zero.

The *dimension* of any set $E$ is the unique number $s$ such that $\Lambda^t E = 0$ for any $t > s$, and $\Lambda^t E = \infty$ for any $t < s$.

1.2. Chapter I contains some preliminary lemmas which enable us subsequently to prove the main theorems of this paper.

In Chapter II we prove that any $s$-set projects into a (linear) set of positive Lebesgue measure in almost all directions if $s > 1$, and into a (linear) set of dimension $s$ if $s \leqslant 1$. In addition we prove that almost every line through almost every point of any $s$-set intersects it in a set of dimension $s-1$ if $s > 1$. We deduce some related theorems, and construct some examples to illustrate the general theory.

Chapter III is concerned entirely with local properties of $s$-sets. First we extend to $s$-sets the definitions of angular density given by Besicovitch (**2**)

for linearly measurable sets, and then prove some new theorems concerning both upper and lower angular density. Finally, using one of these theorems, we prove the non-existence of regular $s$-sets for values of $s$ other than 0, 1, or 2.

## CHAPTER I

## 2. Elementary lemmas

2.1. NOTATION AND DEFINITIONS. Given any point $a$ in the plane, and any positive number $q$, we shall denote by $U(a,q)$ any convex area $U$ such that

$$a \in U \quad \text{and} \quad dU < q.$$

Suppose that $E$ is any $s$-set. Then by

$$\Gamma^s_q(E,a)$$

we shall denote the upper bound of

$$\frac{\Lambda^s E \,.\, U(a,q)}{[dU(a,q)]^s},$$

taken over all possible areas $U(a,q)$.

The *convex density* of $E$ at $a$ is to be given by

$$\Gamma^s(E,a) = \lim_{q \to 0} \Gamma^s_q(E,a).$$

By means of straightforward generalizations of the proofs of the theorems of Besicovitch ((2), 296–306) it is not difficult to prove

LEMMA 1. *The convex density of any $s$-set is equal to unity at almost all its points, and equal to zero at almost all points outside it.*

2.2. NOTATION AND DEFINITIONS. By $c(a,r)$ is denoted the closed circle with centre at the point $a$ and radius $r$. The *upper and lower circular densities* of an $s$-set $E$ at $a$ are given by

$$\overline{D}^s(E,a) = \overline{\lim_{r \to 0}} \frac{\Lambda^s E \,.\, c(a,r)}{2^s r^s} \quad \text{and} \quad \underline{D}^s(E,a) = \underline{\lim_{r \to 0}} \frac{\Lambda^s E \,.\, c(a,r)}{2^s r^s}.$$

If these densities are equal to each other, their common value is called the *circular density*.

It follows from our definitions that if $E$ is any $s$-set, and $a$ is an arbitrary point in the plane, then

$$2^{-s}\Gamma^s(E,a) \leqslant \overline{D}^s(E,a) \leqslant \Gamma^s(E,a).$$

Using this, we are able to deduce the following three lemmas from Lemma 1.

LEMMA 2. *The upper circular density of any $s$-set is less than or equal to unity at almost all its points.*

LEMMA 3. *The upper circular density of any s-set is greater than or equal to* $2^{-s}$ *at almost all its points.*

LEMMA 4. *At almost all points outside any s-set, the circular density is equal to zero.*

2.3. The following lemma can be proved by the same method as Besicovitch and Moran (**5**) used for the case of linear sets.

LEMMA 5. *Any s-set E contains a closed set whose measure is arbitrarily near to that of E.*

## 3. Fubini's theorem

3.1. Suppose that $E$ is any $s$-set in the plane $O_{xy}$ and that $Z$ is any Lebesgue measurable set on the axis $O_z$. Denote by $\mathfrak{E}$ the class of all $s$-sets $E_1 \subset E$ and by $\mathfrak{Z}$ the class of all Lebesgue measurable sets $Z_1 \subset Z$. Thus $\mathfrak{E}$ and $\mathfrak{Z}$ are additive classes of sets in the spaces $E$ and $Z$ respectively. We shall denote by $\mathfrak{E}\mathfrak{Z}$ the smallest additive class in the cartesian product space $E \times Z$ that contains all product sets of the form $E_1 \times Z_1$, where $E_1 \in E$ and $Z_1 \in Z$ (see Saks (**14**), § 9, 82).

LEMMA 6. *If B is a Borel set in* $R^3$, *then in the above notation,*

$$B.(E \times Z) \in \mathfrak{E}\mathfrak{Z}.$$

*Proof.* Denote by $\mathfrak{A}$ the class of all sets $A$ in $R^3$ which satisfy

$$A.(E \times Z) \in \mathfrak{E}\mathfrak{Z}.$$

Clearly, $\mathfrak{A}$ is an additive class in $R^3$.

Suppose that $C$ is any open cube. Then we can write

$$C = S \times I,$$

where $S$ is an open square in $O_{xy}$ and $I$ is an interval on $O_z$. We have

$$SE \in \mathfrak{E} \quad \text{and} \quad IZ \in \mathfrak{Z},$$

and therefore $$C.(E \times Z) = SE \times IZ \in \mathfrak{E}\mathfrak{Z}.$$

That is to say, $C \in \mathfrak{A}$. It follows that $\mathfrak{A}$ contains all Borel sets, and the proof is complete.

3.2. NOTATION AND DEFINITION. We shall call a single-valued function $f(x, y)$ *s-measurable* if, for every number $\alpha$, the set of points $(x, y)$ such that $f(x, y) > \alpha$ is $s$-measurable. If, in addition, $f(x, y)$ is non-negative, we shall denote its integral with respect to $s$-dimensional measure, taken over a given $s$-set $E$, by

$$\int_E f(x, y)\, d\Lambda^s(x, y).$$

By virtue of Lemma 6, we are now able to state the following lemma as a consequence of the abstract form of Fubini's theorem in Saks (**14**) (§ 9, 85).

LEMMA 7 (the Fubini lemma). *Suppose that $f(x,y,z)$ is any non-negative Borel measurable function, that $E$ is any $s$-set in the plane $O_{xy}$, and that $Z$ is any Lebesgue measurable set on the axis $O_z$. Then*

$$\int_Z dz \int_E f(x,y,z)\, d\Lambda^s(x,y) = \int_E d\Lambda^s(x,y) \int_Z f(x,y,z)\, dz,$$

*and all the integrands are (appropriately) measurable functions.*

## 4. Fundamental lemmas

4.1. In order to simplify the proofs of lemmas and theorems, we shall assume that neither the circumference of any circle nor any straight line intersects any $s$-set under consideration in a set of positive $s$-dimensional measure. This assumption is in any case justified if $s > 1$. Furthermore, Theorem II, which is our main theorem concerning sets of dimension not greater than unity, is trivial if this is not the case.

4.2. NOTATION AND DEFINITIONS. By the *direction* $\theta$ we mean the direction in the plane $O_{xy}$ which makes an angle $\theta$ with the positive direction of the $x$-axis.

By $S(a;\theta,\delta,r)$ we denote the closed section of the circle $c(a,r)$ cut off by the two parallel lines which lie in the direction $\theta$ and whose distances from $a$ are equal to $\delta$, where $\delta < r$.

By $R(a;\theta,\delta,d)$ we denote the closed rectangle whose centre is at $a$, whose base is of length $2\delta$, and whose sides are of length $2d$ and lie in the direction $\theta$.

The latter notation will frequently be replaced by $R(x,y;\theta,\delta,d)$, where $x$ and $y$ are the coordinates of $a$. This method of modifying notation will also be applied to other geometrical figures whose position is dependent upon a point $a$.

Suppose that $\Psi$ is a set of angles whose principal values are represented as a set of points in the interval $[0, 2\pi]$. Then the *measure* of $\Psi$ is defined as the linear measure of this set of points, and is denoted by $|\Psi|$.

LEMMA 8. *Given any $s$-set $E$, any point $a$, and any positive numbers $\delta$ and $r$, $\delta < r$, then*

$$\int_0^{2\pi} \Lambda^s E \,.\, S(a;\theta,\delta,r)\, d\theta < 2\pi\delta\left[r^{-1}\Lambda^s E \,.\, c(a,r) + \int_\delta^r \rho^{-2}\Lambda^s E \,.\, c(a,\rho)\, d\rho\right], \quad (1)$$

*and the integrands are Lebesgue measurable functions of $\theta$ and $r$ respectively.*

The measurability of the integrands will be apparent in the proof.

*Proof.* Define the function $f(x,y,\theta)$ as the characteristic function of the set of values of the variables $x$, $y$ and $\theta$ such that

$$(x,y) \in S(a;\theta,\delta,r). \quad (2)$$

The set of points $(x, y, \theta)$ in $R^3$ which correspond to these values of the variables is a closed set, and hence our function is Borel measurable. Applying Lemma 7 (the Fubini lemma), we have

$$\int_0^{2\pi} \Lambda^s[E.S(a;\theta,\delta,r)-E.c(a,\delta)]\,d\theta = \int_0^{2\pi} d\theta \int_{E.c(a,r)-E.c(a,\delta)} f(x,y,\theta)\,d\Lambda^s(x,y)$$

$$= \int_{E.c(a,r)-E.c(a,\delta)} d\Lambda^s(x,y) \int_0^{2\pi} f(x,y,\theta)\,d\theta$$

$$= \int_{E.c(a,r)-E.c(a,\delta)} |\Psi(x,y)|\,d\Lambda^s(x,y), \quad (3)$$

where $\Psi(x,y)$, dependent upon the point $(x,y)$, is the set of angles $\theta$ such that (2) holds.

Suppose that $(x,y)$ is any point in $E.c(a,r)-E.c(a,\delta)$, and that

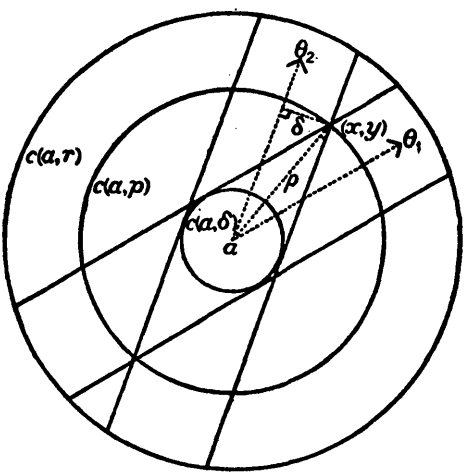



FIG. 1

$\rho = \rho(x,y)$ is its distance from $a$. Choose angles $\theta_1$ and $\theta_2$ so that $(x,y)$ lies on the left-hand side of $S(a;\theta_1,\delta,r)$ and on the right-hand side of $S(a;\theta_2,\delta,r)$ (as in Fig. 1). Thus our set $\Psi(x,y)$ consists of the set of angles $\theta$ between $\theta_1$ and $\theta_2$ together with the diametrically opposite set. By elementary geometry,

$$|\Psi(x,y)| = 2(\theta_2-\theta_1) = 4\sin^{-1}\frac{\delta}{\rho} < 2\pi\frac{\delta}{\rho},$$

whence from (3),

$$\int_0^{2\pi} \Lambda^s[E.S(a;\theta,\delta,r)-E.c(a,\delta)]\,d\theta < 2\pi\delta \int_{E.c(a,r)-E.c(a,\delta)} \rho^{-1}\,d\Lambda^s(x,y). \quad (4)$$

Suppose that $n$ is an arbitrary positive integer, and that the range $(\delta, r)$

is divided into $n-1$ equal parts by the values $\rho = \rho_i$, $i = 1, 2, 3, \ldots, n$, where $\rho_1 = \delta$ and $\rho_n = r$. We have

$$\sum_{i=1}^{n-1} \rho_{i+1}^{-1} \Lambda^s[E.c(a,\rho_{i+1}) - E.c(a,\rho_i)]$$
$$< \int_{E.c(a,r)-E.c(a,\delta)} \rho^{-1}\, d\Lambda^s(x,y) < \sum_{i=1}^{n-1} \rho_i^{-1} \Lambda^s[E.c(a,\rho_{i+1}) - E.c(a,\rho_i)].$$

The common limit of these two sums, as $n \to \infty$, is a Riemann–Stieltjes integral, and so

$$\int_{E.c(a,r)-E.c(a.\delta)} \rho^{-1}\, d\Lambda^s(x,y) = \int_{\rho=\delta}^{r} \rho^{-1}\, d\Lambda^s E.c(a,\rho). \tag{5}$$

Since the boundary of every circle $c(a,\rho)$ intersects $E$ in a set of measure zero, it follows that $\Lambda^s E.c(a,\rho)$ is a continuous function of $\rho$. Integrating the right-hand side of (5) by parts,

$$\int_{E.c(a,r)-E.c(a,\delta)} \rho^{-1}\, d\Lambda^s(x,y)$$
$$= r^{-1}\Lambda^s E.c(a,r) - \delta^{-1}\Lambda^s E.c(a,\delta) + \int_{\delta}^{r} \rho^{-2}\Lambda^s E.c(a,\rho)\, d\rho. \tag{6}$$

Finally we observe that

$$\int_0^{2\pi} \Lambda^s E.S(a;\theta,\delta,r)\, d\theta = 2\pi\Lambda^s E.c(a,\delta) + \int_0^{2\pi} \Lambda^s[E.S(a;\theta,\delta,r) - E.c(a,\delta)]\, d\theta. \tag{7}$$

Combining (4), (6), and (7), we obtain the inequality (1).

LEMMA 9. *At almost all points $a$ of any $s$-set $E$, for all sufficiently small positive numbers $\delta$ and $d$ such that $\delta < d$,*

$$\int_0^{2\pi} \Lambda^s E.R(a;\theta,\delta,d)\, d\theta < K_s d^{s-1}\delta \quad (s > 1), \tag{8}$$
$$< K_s \delta\left(1 + \log\frac{d}{\delta}\right) \quad (s = 1), \tag{9}$$
$$< K_s \delta^s \quad (s < 1), \tag{10}$$

*where $K_s$ is a finite numerical function of $s$ alone.*

*Proof*, By Lemma 2, at almost all points of $E$, for all sufficiently small positive values of $\rho$, $\Lambda^s E.c(a,\rho) < 2^{s+1}\rho^s$.

Therefore for all sufficiently small positive values of $r$, and any positive value of $\delta$ such that $\delta < r$,

$$\int_{\delta}^{r} \rho^{-2}\Lambda^s E.c(a,\rho)\, d\rho < 2^{s+1} \int_{\delta}^{r} \rho^{s-2}\, d\rho \tag{11}$$

and

$$\Lambda^s E.c(a,r) < 2^{s+1} r^s. \tag{12}$$

Setting $r = 2d$, and using (11) and (12) in (1) of Lemma 8, we have

$$\int_0^{2\pi} \Lambda^s E . S(a;\theta,\delta,2d)\, d\theta < 2^{s+2}\pi\delta\left[2^{s-1}d^{s-1} + \int_\delta^{2d} \rho^{s-2}\, d\rho\right]. \tag{13}$$

Now since $\delta < d$, it follows that, for every value of $\theta$,

$$S(a;\theta,\delta,2d) \supset R(a;\theta,\delta,d),$$

and, consequently, in (13) we may replace $S(a;\theta,\delta,2d)$ by $R(a;\theta,\delta,d)$.

The inequalities (8), (9), and (10) can now be seen by integration.

4.3. LEMMA 10. *Suppose that for every value of $\lambda$, $f(\lambda,\theta)$ is a non-negative measurable function of $\theta$, that $p$ and $q$ are positive numbers, and that*

(i) *for every value of $\theta$, $\lambda^p f(\lambda,\theta)$ is a non-decreasing function of $\lambda$,*

(ii) $\int_0^{2\pi} f(\lambda,\theta)\, d\theta = O(\lambda^q)$ *as* $\lambda \to +0$.

*Then*

$$\int_0^{2\pi} \overline{\lim_{\lambda\to+0}} f(\lambda,\theta)\, d\theta = 0.$$

*Proof.* From (ii) we can find a positive number $M$ and a positive integer $n_0$ such that

$$\int_0^{2\pi} f(2^{-n},\theta)\, d\theta < 2^{-nq}M \quad \text{whenever } n > n_0.$$

Thus

$$\int_0^{2\pi} \sum_{n=1}^{\infty} f(2^{-n},\theta)\, d\theta = \sum_{n=1}^{\infty} \int_0^{2\pi} f(2^{-n},\theta)\, d\theta < \infty.$$

It follows that, for almost all values of $\theta$, the integrand of the left-hand side is finite, and in particular that

$$\lim_{n\to\infty} f(2^{-n},\theta) = 0. \tag{14}$$

Suppose now that $\lambda$ is any number such that $0 < \lambda < 1$, and denote by $n$ the integer for which $2^{-(n+1)} \leqslant \lambda < 2^{-n}$. It follows from (i) that for every value of $\theta$,

$$2^{-(n+1)p} f(\lambda,\theta) \leqslant \lambda^p f(\lambda,\theta) \leqslant 2^{-np} f(2^{-n},\theta),$$

whence

$$f(\lambda,\theta) \leqslant 2^p f(2^{-n},\theta).$$

This enables us to deduce from (14) that for almost all values of $\theta$,

$$\lim_{\lambda\to+0} f(\lambda,\theta) = 0,$$

and the lemma is proved.

CHAPTER II

## 5. Orthogonal projections of plane sets

5.1. In the proofs of subsequent lemmas and theorems we shall assume the (appropriate) measurability of certain sets and functions related to the function $f(x, y, \theta, \delta, d) = \Lambda^s E \,.\, R(x, y; \theta, \delta, d)$. To fix the ideas, the following lemma is proved as a sample.

LEMMA 11. *Given positive numbers $p$ and $q$, and an $s$-set $E$, the function*

$$\overline{\lim_{d \to 0}} \varliminf_{\delta \to 0} \delta^{-p} d^{-q} \Lambda^s E \,.\, R(x, y; \theta, \delta, d)$$

*is a Borel measurable function of the variables $x$, $y$, and $\theta$.*

*Proof.* We show first that $\Lambda^s E \,.\, R(x, y; \theta, \delta, d)$ is a continuous function of $x$, $y$ and $\theta$.

Suppose that $x_0$, $y_0$ and $\theta_0$ are fixed values of the variables, and that $\epsilon$ is an arbitrary positive number. Then for all values of $x, y$ and $\theta$ sufficiently near to $x_0$, $y_0$ and $\theta_0$, the points which are not common to both

$$E \,.\, R(x, y; \theta, \delta, d) \quad \text{and} \quad E \,.\, R(x_0, y_0; \theta_0, \delta, d)$$

lie inside the set

$$E[R(x_0, y_0; \theta_0, \delta+\epsilon, d+\epsilon) - R(x_0, y_0; \theta_0, \delta-\epsilon, d-\epsilon)],$$

and consequently

$$|\Lambda^s E \,.\, R(x, y; \theta, \delta, d) - \Lambda^s E \,.\, R(x_0, y_0; \theta_0, \delta, d)|$$
$$\leqslant \Lambda^s E[R(x_0, y_0; \theta_0, \delta+\epsilon, d+\epsilon) - R(x_0, y_0; \theta_0, \delta-\epsilon, d-\epsilon)].$$

Since the boundary of $R(x_0, y_0; \theta_0, \delta, d)$ intersects $E$ in a set of measure zero, the right-hand side tends to zero as $\epsilon \to 0$, and the continuity of our function follows.

To complete the proof, we make use of the fact that if $f(\alpha)$ is any non-negative, non-decreasing function of $\alpha$, then

$$\varliminf_{\alpha \to +0} \alpha^{-p} f(\alpha) = \varliminf_{n \to \infty} n^p f(n^{-1}) \quad \text{and} \quad \overline{\lim_{\alpha \to +0}} \alpha^{-q} f(\alpha) = \overline{\lim_{n \to \infty}} n^q f(n^{-1}),$$

where $n$ takes only integral values.

Setting $\alpha = \delta$, the first equality gives

$$\varliminf_{\delta \to 0} \delta^{-p} d^{-q} \Lambda^s E \,.\, R(x, y; \theta, \delta, d) = \varliminf_{n \to \infty} n^p d^{-q} \Lambda^s E \,.\, R(x, y; \theta, n^{-1}, d),$$

and the Borel measurability of the function on the left-hand side follows. Setting $\alpha = d$, and applying the same argument in turn to this function, the Borel measurability of the function of the lemma is proved.

Since $$\varliminf_{\delta \to 0} \delta^{-p} \Lambda^s E \,.\, R(x, y; \theta, \delta, d)$$

is a non-decreasing function of $d$, it follows that

$$\lim_{d\to 0}\varlimsup_{\delta\to 0}\delta^{-p}\Lambda^s E.\,R(x,y;\theta,\delta,d)$$

exists. Similarly, so does

$$\lim_{d\to 0}\overline{\lim_{\delta\to 0}}\,\delta^{-p}\Lambda^s E.\,R(x,y;\theta,\delta,d).$$

These facts will always be understood in the proofs.

5.2. NOTATION. By $\mathrm{proj}_\theta E$ is denoted the (linear) set obtained by orthogonal projection of the set $E$ onto a line which lies in the direction $\theta+\frac{1}{2}\pi$.

LEMMA 12. *If $E$ is an $s$-set and $\theta$ an angle such that $|\mathrm{proj}_\theta E| = 0$, then at almost all points $(x,y)$ of $E$ we have*

$$\lim_{d\to 0}\varlimsup_{\delta\to 0}\delta^{-1}\Lambda^s E.\,R(x,y;\theta,\delta,d) = \infty.$$

*Proof.* For convenience, we set $\theta = \frac{1}{2}\pi$ and represent $\mathrm{proj}_\theta E$ on the $x$-axis.

Suppose that $\alpha$ and $d$ are arbitrary positive numbers and that $n$ is an integer, and denote by $E_n = E_n(\alpha, d)$ the set of points $(x,y)$ of $E$ such that

$$\varlimsup_{\delta\to 0}\delta^{-1}\Lambda^s E.\,R(x,y;\theta,\delta,d) < \alpha \tag{1}$$

and

$$nd \leqslant y \leqslant (n+1)d. \tag{2}$$

By Lemma 5 we can find a closed set $F_n \subset E_n$ which satisfies the inequality

$$\Lambda^s F_n \geqslant \tfrac{1}{2}\Lambda^s E_n. \tag{3}$$

Since $|\mathrm{proj}_\theta F_n| = 0$, we can, given $\epsilon > 0$, find a sequence of closed intervals $I_u$, $u = 1, 2, 3, \ldots$, such that

$$\sum_{u=1}^{\infty} I_u \supset \mathrm{proj}_\theta F_n \tag{4}$$

and

$$\sum_{u=1}^{\infty} |I_u| < \epsilon. \tag{5}$$

Now, given any closed interval $I$ on the $x$-axis, let us denote by $R_n(I)$ the closed rectangle which is the cartesian product set of $I$ and the interval $[nd, (n+1)d]$ on the $y$-axis.

In this notation (4) becomes

$$\sum_{u=1}^{\infty} R_n(I_u) \supset F_n,$$

whence

$$\sum_{u=1}^{\infty} \Lambda^s F_n.\,R_n(I_u) \geqslant \Lambda^s F_n. \tag{6}$$

On the other hand, by means of (1) we shall show that for all integers $u$,

$$|I_u|^{-1}\Lambda^s F_n.\,R_n(I_u) < 2\alpha. \tag{7}$$

For, suppose that for some integer $u$ (7) does not hold, and denote $I_u$ simply by $J$. Thus

$$|J|^{-1}\Lambda^s F_n . R_n(J) \geqslant 2\alpha.$$

Divide $J$ into two equal closed intervals. Then at least one of these, $J_1$ say, will satisfy

$$|J_1|^{-1}\Lambda^s F_n . R_n(J_1) \geqslant 2\alpha.$$

Similarly divide $J_1$ into two equal closed intervals and denote by $J_2$ the one which satisfies

$$|J_2|^{-1}\Lambda^s F_n . R_n(J_2) \geqslant 2\alpha.$$

Proceed indefinitely in this way, so that we form a converging sequence of intervals $J_i$, $i = 1, 2, 3,...$, and for every one,

$$|J_i| = 2^{-i}|J| \tag{8}$$

and

$$|J_i|^{-1}\Lambda^s F_n . R_n(J_i) \geqslant 2\alpha. \tag{9}$$

Since $F_n$ is closed, and the sets $F_n . R_n(J_i)$, $i = 1, 2, 3,...$, are not empty, we can find a point

$$(x, y) \in \prod_{i=1}^{\infty} F_n . R_n(J_i).$$

Furthermore for every integer $i$ we have

$$R(x, y; \theta, |J_i|, d) \supset R_n(J_i),$$

and hence from (8) and (9),

$$2^i|J|^{-1}\Lambda^s F_n . R(x, y; \theta, 2^{-i}|J|, d) \geqslant 2\alpha.$$

Therefore

$$\varlimsup_{\delta \to 0} \delta^{-1}\Lambda^s F_n . R(x, y; \theta, \delta, d) \geqslant \alpha.$$

Since $(x, y) \in F_n$ and $F_n \subset E_n$, this is in contradiction with (1), and so the inequality (7) must hold for all integers $u$. Together with (5) this gives

$$\sum_{u=1}^{\infty} \Lambda^s F_n . R_n(I_u) < 2\alpha\epsilon,$$

whence, from (6),

$$\Lambda^s F_n < 2\alpha\epsilon.$$

By making $\epsilon \to 0$, we infer that $\Lambda^s F_n = 0$ and hence from (3) that $\Lambda^s E_n = 0$. Summing $\Lambda^s E_n$ over all the integers $n$, we deduce from (1) and (2) that at almost all points $(x, y)$ of $E$,

$$\varlimsup_{\delta \to 0} \delta^{-1}\Lambda^s E . R(x, y; \theta, \delta, d) \geqslant \alpha.$$

The lemma now follows because $\alpha$ and $d$ are arbitrary.

5.3. In this paragraph we prove a lemma which will have several applications later on, and then we deduce our first theorem.

NOTATION. In future, for $R(x, y; \theta, \delta, d)$ we shall often simply write $R$.

LEMMA 13. *If $E$ is an $s$-set and $s > 1$, then for almost all angles $\theta$, all $s$-sets $A$ which are contained in $E$ satisfy* $|\mathrm{proj}_\theta A| > 0$.

*Proof.* By Lemma 9, at almost all points $(x, y)$ of $E$, for all sufficiently small positive numbers $d$ we have

$$\overline{\lim_{\delta \to 0}} \int_0^{2\pi} \delta^{-1} \Lambda^s ER \, d\theta < K_s d^{s-1},$$

and consequently, by Fatou's theorem,

$$\int_0^{2\pi} \overline{\lim_{\delta \to 0}} \delta^{-1} \Lambda^s ER \, d\theta < K_s d^{s-1}.$$

By making $d \to 0$ and applying Fatou's theorem again, we have

$$\int_0^{2\pi} \lim_{d \to 0} \overline{\lim_{\delta \to 0}} \delta^{-1} \Lambda^s ER \, d\theta = 0.$$

By Lemma 11 the above integrand is a Borel measurable function of $x$, $y$ and $\theta$, and so, by the Fubini lemma,

$$\int_0^{2\pi} d\theta \int_E \lim_{d \to 0} \overline{\lim_{\delta \to 0}} \delta^{-1} \Lambda^s ER \, d\Lambda^s(x, y) = \int_E d\Lambda^s(x, y) \int_0^{2\pi} \lim_{d \to 0} \overline{\lim_{\delta \to 0}} \delta^{-1} \Lambda^s ER \, d\theta = 0. \tag{10}$$

On the other hand, for every angle $\theta$ which does not satisfy the condition required by this lemma, we have by Lemma 12,

$$\int_0^{2\pi} \lim_{d \to 0} \overline{\lim_{\delta \to 0}} \delta^{-1} \Lambda^s ER \, d\Lambda^s(x, y) = \infty.$$

The substitution of this into (10) is only possible if the set of all such angles $\theta$ is of measure zero, and so the proof is complete.

We deduce immediately

THEOREM I. *Any s-set whose dimension is greater than unity projects into a set of positive Lebesgue measure in almost all directions.*

5.4. In this paragraph we prove a lemma analogous to Lemma 12.

LEMMA 14. *If $E$ is an s-set, $t$ a positive number, and $\theta$ an angle such that $\Lambda^t \operatorname{proj}_\theta E = 0$, then at almost all points $(x, y)$ of $E$,*

$$\lim_{d \to 0} \overline{\lim_{\delta \to 0}} \delta^{-t} \Lambda^s ER = \infty.$$

*Proof.* For convenience, we set $\theta = \frac{1}{2}\pi$ and represent $\operatorname{proj}_\theta E$ on the $x$-axis.

Suppose that $\alpha$, $\eta$, and $d$ are arbitrary positive numbers and that $n$ is an arbitrary integer, and denote by $E_n = E_n(\alpha, \eta, d)$ the set of points $(x, y)$ of $E$ such that

$$\sup_{\delta < \eta} \delta^{-t} \Lambda^s ER < \alpha \tag{11}$$

and

$$nd \leqslant y \leqslant (n+1)d. \tag{12}$$

Since $\Lambda^t \mathrm{proj}_\theta E_n = 0$, we can, given $\epsilon > 0$, find a sequence of closed intervals $I_u$, $u = 1, 2, 3,...$, such that

$$\sum_{u=1}^{\infty} I_u \supset \mathrm{proj}_\theta E_n, \tag{13}$$

$$\sum_{u=1}^{\infty} |I_u|^t < \epsilon, \tag{14}$$

and

$$|I_u| < \eta, \quad u = 1, 2, 3,.... \tag{15}$$

Given any closed interval $I$ on the $x$-axis, we shall, as in the proof of Lemma 12, denote by $R_n(I)$ the rectangle $I \times [nd, (n+1)d]$.

In this notation (13) becomes

$$\sum_{u=1}^{\infty} R_n(I_u) \supset E_n,$$

whence

$$\sum_{u=1}^{\infty} \Lambda^s E_n . R_n(I_u) \geqslant \Lambda^s E_n. \tag{16}$$

On the other hand, from (11) and (15),

$$|I_u|^{-t} \Lambda^s E_n . R_n(I_u) < \alpha, \quad u = 1, 2, 3,...,$$

and together with (14) this gives

$$\sum_{u=1}^{\infty} \Lambda^s E_n . R_n(I_u) < \alpha\epsilon.$$

Therefore from (16) we have $\Lambda^s E_n < \alpha\epsilon$. By making $\epsilon \to 0$ we see that $\Lambda^s E_n = 0$, and hence that

$$\sum_{n=-\infty}^{\infty} \Lambda^s E_n = 0.$$

We deduce from (11) and (12) that at almost all points $(x, y)$ of $E$,

$$\sup_{\delta < \eta} \delta^{-t} \Lambda^s ER \geqslant \alpha.$$

The lemma now follows because $\alpha$, $\eta$, and $d$ are arbitrary.

5.5. We shall now prove the analogue of Theorem I.

THEOREM II. *Any $s$-set whose dimension does not exceed unity projects into a set of dimension $s$ in almost all directions.*

*Proof.* It will be understood throughout the proof that $s$ is a fixed positive number such that $s \leqslant 1$. Suppose then that $E$ is any $s$-set, and suppose that $t$ is any positive number such that $t < s$. It follows from Lemma 9 that at almost all points $(x, y)$ of $E$, for all sufficiently small positive numbers $d$ and any positive number $\delta$ such that $\delta < d$,

$$\int_0^{2\pi} \delta^{-t} \Lambda^s ER \, d\theta < K_s \delta^{s-t} \left(1 + \log \frac{d}{\delta}\right) = o\left(\delta^{\frac{1}{2}(s-t)}\right) \quad \text{as } \delta \to 0.$$

Denote the integrand of this inequality by $f(\delta, \theta)$. Then $\delta^t f(\delta, \theta)$ is a non-decreasing function of $\delta$. In addition, $s-t$ is positive, and so by Lemma 10

$$\int_0^{2\pi} \overline{\lim_{\delta \to 0}}\, \delta^{-t} \Lambda^s ER \, d\theta = 0.$$

Since this holds for all sufficiently small positive values of $d$, it follows by Fatou's theorem that at almost all points $(x, y)$ of $E$,

$$\int_0^{2\pi} \lim_{d \to 0} \overline{\lim_{\delta \to 0}}\, \delta^{-t} \Lambda^s ER \, d\theta = 0.$$

By applying the Fubini lemma, and using Lemma 14, we deduce that for almost all angles $\theta$, $\quad \Lambda^t \mathrm{proj}_\theta E > 0.$

Thus for almost all angles $\theta$ we have

$$\Lambda^t \mathrm{proj}_\theta E > 0 \quad \text{whenever } 0 \leqslant t < s,$$

and for all angles $\theta$, $\quad \Lambda^s \mathrm{proj}_\theta E \leqslant \Lambda^s E < \infty.$

This completes the proof.

5.6. Remarks. Theorem II leaves open the question as to whether the projections are of positive or zero $s$-dimensional measure. The first possibility is easily illustrated by considering any plane $s$-set which is the subset of a straight line. Now Besicovitch ((1), 456–8) has constructed a linearly measurable set which projects into a set of measure zero in all directions (see also Gillis (10)). By modifying sets of this type, we can construct $s$-sets, where $0 < s < 1$, which project into sets of zero $s$-dimensional measure in all directions, and these illustrate the second possibility.

It has been shown by Davies (6) that any analytic set of infinite $s$-dimensional measure contains a closed $s$-set. The two theorems we have proved therefore hold for all analytic sets as well as for $s$-sets.

Besicovitch (3) has proved that every irregular linearly measurable set projects into a set of zero Lebesgue measure in almost all directions (see also Federer (9) where the result is generalized to $n$-dimensional space). Theorem II shows that for almost all directions this set is of dimension unity, and Theorem I shows that the result is never true for $s$-sets of dimension greater than unity.

## 6. Intersections of plane sets with straight lines

6.1. Notation. By $(a, \theta)$ we shall denote the infinite segment (or ray) through the point $a$, lying in the direction $\theta$, and with $a$ as end-point, but not actually including the point $a$.

Lemma 15. *If $F$ is a closed set, and $t$ is a positive number, then $\Lambda^t(x, y; \theta).F$ is a Borel measurable function of $x$, $y$ and $\theta$.*

*Proof.* We can assume without loss of generality that $F$ is a bounded set.

Suppose that $q$ and $\alpha$ are arbitrary positive numbers, and that $x_0$, $y_0$ and $\theta_0$ are any fixed values of the variables such that

$$\Lambda_q^t(x_0, y_0; \theta_0).F < \alpha.$$

Choose a finite sequence of open intervals $K_j, j = 1, 2, 3, \ldots, n$, on $(x_0, y_0; \theta_0)$ such that

$$\sum_{j=1}^{n} K_j \supset (x_0, y_0; \theta_0).F, \tag{1}$$

$$\sum_{j=1}^{n} |K_j|^t < \alpha, \tag{2}$$

and

$$|K_j| < q, \quad j = 1, 2, 3, \ldots, n. \tag{3}$$

For each interval $K_j$ denote by $A(K_j)$ the open annulus whose centre is at $(x_0, y_0)$ and whose boundaries pass through the end-points of $K_j$. (If one of the intervals $K_j$ has $(x_0, y_0)$ as end-point, then $A(K_j)$ is taken as the open circle whose boundary passes through the other end-point.) Since $F$ is closed and bounded, and the intervals $K_j$ are open and finite in number, it is evident from (1) that for all values of $x$, $y$ and $\theta$ sufficiently near to $x_0$, $y_0$ and $\theta_0$,

$$\sum_{j=1}^{n} A(K_j) \supset (x, y; \theta).F. \tag{4}$$

For each annulus $A(K_j)$ denote by $K_j' = K_j'(x, y, \theta)$ the open interval $(x, y; \theta).A(K_j)$. In this notation (4) becomes

$$\sum_{j=1}^{n} K_j' \supset (x, y; \theta).F.$$

Moreover the quantities $|K_j'|, j = 1, 2, 3, \ldots, n$, are all continuous functions of $x$, $y$ and $\theta$. It follows that, provided $x$, $y$ and $\theta$ are sufficiently near to $x_0$, $y_0$ and $\theta_0$, then (1), (2), and (3) hold with $K_j$ replaced by $K_j'$ in each case. Consequently for all such values of $x$, $y$ and $\theta$,

$$\Lambda_q^t(x, y; \theta).F < \alpha.$$

Since $\alpha$ is arbitrary, we deduce that the function $\Lambda_q^t(x, y; \theta).F$ is upper semi-continuous. In addition it is a non-increasing function of $q$ and so the Borel measurability of

$$\lim_{q \to 0} \Lambda_q^t(x, y; \theta).F = \Lambda^t(x, y; \theta).F$$

follows.

6.2. We next prove a lemma of the same type as Lemmas 12 and 14.

LEMMA 16. *If $E$ is an $s$-set, $t$ a positive number, and $\theta$ an angle such that for every line $l$ which lies in the direction $\theta$, $\Lambda^t E.l = 0$, then at almost all points $(x, y)$ of $E$*

$$\overline{\lim_{d \to 0}} \lim_{\delta \to 0} \delta^{-1} d^{-t} \Lambda^s E R = \infty.$$

*Proof.* We can assume without loss of generality that $E$ is a bounded set.

For convenience, set $\theta = \frac{1}{2}\pi$ and assume that $E$ lies between the lines $x = 0$ and $x = 1$.

Suppose that $\alpha$ and $\kappa$ are arbitrary positive numbers, and denote by $E_1' = E_1(\alpha, \kappa)$ the set of points $(x, y)$ of $E$ such that

$$\sup_{d<\kappa} \varlimsup_{\delta\to 0} \delta^{-1} d^{-t} \Lambda^s E R < \alpha. \tag{5}$$

Choose a closed set $F_1 \subset E_1$ such that

$$\Lambda^s F_1 \geqslant \tfrac{1}{2}\Lambda^s E_1. \tag{6}$$

Given any interval $I$ on the $x$-axis, we shall denote by $S(I)$ the cartesian product of $I$ and the $y$-axis.

Now there must exist a point $(\xi, 0)$ in the interval $[0, 1]$ and a sequence of closed intervals $I_u$, $u = 1, 2, 3, \ldots$, such that

$$(\xi, 0) \in I_u, \quad u = 1, 2, 3, \ldots,$$

$$\lim_{u\to\infty} |I_u| = 0, \tag{7}$$

and

$$|I_u|^{-1} \Lambda^s F_1 . S(I_u) \geqslant \Lambda^s F_1, \quad u = 1, 2, 3, \ldots. \tag{8}$$

Denote the line $x = \xi$ by $l$. The set $F_1 . l$ is closed and bounded, and $\Lambda^t F_1 . l = 0$, and so we can, given $\epsilon > 0$, find a finite sequence of open intervals $K_j$, $j = 1, 2, 3, \ldots, n$, on $l$ such that

$$\sum_{j=1}^{n} K_j \supset F_1 . l, \tag{9}$$

$$\sum_{j=1}^{n} |K_j| < \epsilon, \tag{10}$$

and

$$|K_j| < \kappa, \quad j = 1, 2, 3, \ldots, n. \tag{11}$$

For each interval $K_j$ denote by $T(K_j)$ the cartesian product of $K_j$ and the $x$-axis.

Since $F_1$ is closed, and the intervals $K_j$ are open and finite in number, it is evident from (7) and (9) that for all sufficiently large integers $u$,

$$\sum_{j=1}^{n} T(K_j) . S(I_u) \supset F_1 . S(I_u),$$

whence from (8),

$$\sum_{j=1}^{n} \Lambda^s F_1 . T(K_j) . S(I_u) \geqslant \Lambda^s F_1 . S(I_u) \geqslant |I_u| \Lambda^s F_1. \tag{12}$$

Now each figure $T(K_j) . S(I_u)$ is a rectangle of height $|K_j|$ and base $|I_u|$. Moreover from (5) and (11), for all points $(x, y) \in F_1$ we have

$$\varlimsup_{\delta\to 0} \delta^{-1} \Lambda^s F_1 . R(x, y; \theta, \delta, |K_j|) < \alpha |K_j|^t, \quad j = 1, 2, 3, \ldots, n.$$

By the method of repeatedly subdividing the interval $I_u$ (as was used in the proof of Lemma 12), we can deduce that for all integers $u$,

$$|I_u|^{-t}\Lambda^s F_1 . T(K_j) . S(I_u) < 2\alpha |K_j|^t, \quad j = 1, 2, 3, \ldots, n,$$

whence

$$\sum_{j=1}^{n} \Lambda^s F_1 . T(K_j) . S(I_u) < 2|I_u|\alpha \sum_{j=1}^{n} |K_j|^t$$
$$< 2|I_u|\alpha\epsilon, \quad \text{from (10)}.$$

On the other hand, (12) holds for all sufficiently large integers and so

$$\Lambda^s F_1 < 2\alpha\epsilon.$$

By making $\epsilon \to 0$ we see that $\Lambda^s F_1 = 0$, and it follows from (6) that $\Lambda^s E_1 = 0$. Therefore from (5), at almost all points $(x, y)$ of $E$,

$$\sup_{d<\kappa} \overline{\lim_{\delta \to 0}} \delta^{-1} d^{-t} \Lambda^s E R \geqslant \alpha.$$

The lemma now follows because $\alpha$ and $\kappa$ are arbitrary.

6.3. The following lemma constitutes the fundamental part of the theorem of this section.

LEMMA 17. *Almost every line through almost every point of any $s$-set intersects it in a set whose dimension is at least $s-1$.*

*Proof.* By virtue of Lemma 5 it will be sufficient to prove the lemma for a closed $s$-set $F$, and we can assume, for an obvious reason, that $s > 1$.

Suppose that $t$ is an arbitrary positive number, $t < s-1$. By Lemma 9, at almost all points $(x, y)$ of $F$, for all sufficiently small positive numbers $d$,

$$\overline{\lim_{\delta \to 0}} \int_0^{2\pi} \delta^{-1} d^{-t} \Lambda^s F R \, d\theta < K_s d^{s-t-1},$$

whence by Fatou's theorem,

$$\int_0^{2\pi} \overline{\lim_{\delta \to 0}} \delta^{-1} d^{-t} \Lambda^s F R \, d\theta < K_s d^{s-t-1} = O(d^{s-t-1}) \quad \text{as } d \to 0.$$

Denote the integrand of this inequality by $f(d, \theta)$. Then $d^t f(d, \theta)$ is a non-decreasing function of $d$. In addition, $s-t-1$ is positive, and so by Lemma 10,

$$\int_0^{2\pi} \overline{\lim_{d \to 0}} \overline{\lim_{\delta \to 0}} \delta^{-1} d^{-t} \Lambda^s F R \, d\theta = 0.$$

We now use the Fubini lemma and obtain the equality

$$\int_0^{2\pi} d\theta \int_F \overline{\lim_{d \to 0}} \overline{\lim_{\delta \to 0}} \delta^{-1} d^{-t} \Lambda^s F R \, d\Lambda^s(x, y) = 0. \tag{13}$$

For every angle $\theta$ denote by $E_\theta$ the set of points $(x, y)$ of $F$ such that $\Lambda^t(x, y; \theta).F = 0$. By Lemma 16, at almost all points $(x, y)$ of $E_\theta$,

$$\overline{\lim_{d\to 0}}\,\overline{\lim_{\delta\to 0}}\,\delta^{-1}d^{-t}\Lambda^s FR = \infty,$$

and so it follows from (13) that

$$\int_0^{2\pi} \Lambda^s E_\theta \, d\theta = 0. \tag{14}$$

Define $g(x, y, \theta)$ as the characteristic function of the set of values of $x$, $y$ and $\theta$ such that $\Lambda^t(x, y; \theta).F = 0$. By Lemma 15 this function is Borel measurable and so

$$\int_F d\Lambda^s(x, y) \int_0^{2\pi} g(x, y, \theta)\, d\theta = \int_0^{2\pi} d\theta \int_F g(x, y, \theta)\, d\Lambda^s(x, y) = 0 \quad \text{from (14).}$$

It follows that almost every line through almost every point of $F$ intersects $F$ in a set of positive $t$-dimensional measure. We have proved this true for any positive value of $t$ such that $t < s-1$, and so the proof of the lemma is complete.

6.4. NOTATION. We shall denote the upper Lebesgue integral of any function $f(x)$ by

$$\int^* f(x)\, dx = \inf_{g(x)>f(x)} \int g(x)\, dx,$$

where $g(x)$ can be any Lebesgue measurable function.

By $E_x$ we denote the set of points of $E$ whose abscissae are equal to $x$.

LEMMA 18. *If $E$ is a plane set, $a$ any point in the plane, $r$ a positive number, and $s$ a positive number greater than unity, then*

$$\int_0^{2\pi *} \Lambda^{s-1}(a; \theta).[E-E.c(a, r)]\, d\theta \leqslant r^{-1}\Lambda^s[E-E.c(a, r)]. \tag{15}$$

*Proof.* For every positive integer $n$, choose a sequence of convex areas $U_{n,i}$, $i = 1, 2, 3, \ldots$, such that

$$R^2-c(a, r) \supset \sum_{i=1}^{\infty} U_{n,i} \supset E-E.c(a, r), \tag{16}$$

$$\sum_{i=1}^{\infty} (dU_{n,i})^s < \Lambda^s[E-E.c(a, r)]+n^{-1}, \tag{17}$$

and

$$dU_{n,i} < n^{-1}, \quad i = 1, 2, 3, \ldots. \tag{18}$$

Let $\psi_{n,i}$ denote the angle subtended by $U_{n,i}$ from $a$. Then

$$r \sin \tfrac{1}{2}\psi_{n,i} \leqslant \tfrac{1}{2} dU_{n,i},$$

and hence from (18),

$$\psi_{n,i} \leqslant r^{-1}[1+O(n^{-1})]\, dU_{n,i}.$$

It follows that

$$\int_0^{2\pi}{}^{*} |(a;\theta).U_{n,i}|^{s-1}\,d\theta \leqslant r^{-1}[1+O(n^{-1})]\,(dU_{n,i})^s, \tag{19}$$

and in (17) this gives

$$\int_0^{2\pi} \sum_{i=1}^{\infty} |(a;\theta).U_{n,i}|^{s-1}\,d\theta$$

$$= \sum_{i=1}^{\infty} \int_0^{2\pi} |(a;\theta).U_{n,i}|^{s-1}\,d\theta \leqslant r^{-1}[1+O(n^{-1})]\Lambda^s[E-E.c(a,r)]+O(n^{-1}).$$

By making $n \to \infty$ and using Fatou's theorem, we deduce that

$$\int_0^{2\pi} \varlimsup_{n\to\infty} \sum_{i=1}^{\infty} |(a;\theta).U_{n,i}|^{s-1}\,d\theta \leqslant r^{-1}\Lambda^s[E-E.c(a,r)]. \tag{20}$$

On the other hand, from (16), for every value of $\theta$ and every positive integer $n$,

$$\sum_{i=1}^{\infty} (a;\theta).U_{n,i} \supset (a;\theta).[E-E.c(a,r)],$$

and from (18) $\quad |(a;\theta).U_{n,i}| < n^{-1}, \quad i = 1, 2, 3,\ldots,$

and it follows that

$$\Lambda^{s-1}(a;\theta).[E-E.c(a,r)] \leqslant \varlimsup_{n\to\infty} \sum_{i=1}^{\infty} |(a;\theta).U_{n,i}|^{s-1}. \tag{21}$$

Combining (20) and (21), we obtain the inequality (15).

REMARK. The above lemma is related to the work of Moran (**13**). He obtains a best possible result for an inequality similar to (19).

LEMMA 19. *Almost every line through almost every point of any s-set whose dimension is greater than unity intersects it in a set of finite* $(s-1)$*-dimensional measure.*

*Proof.* Suppose that $a$ is a point in the plane, that $\delta$ and $r$ are positive numbers, $\delta < r$. Divide the range $[\delta, r]$ into $n-1$ equal parts by the values $\rho = \rho_i$, $i = 1, 2, 3,\ldots, n$, where $\rho_1 = \delta$ and $\rho_n = r$.

By Lemma 18

$$\rho_i^{-1}\Lambda^s[E.c(a,\rho_{i+1})-E.c(a,\rho_i)]$$

$$\geqslant \int_0^{2\pi}{}^{*} \Lambda^{s-1}(a;\theta).[E.c(a,\rho_{i+1})-E.c(a,\rho_i)]\,d\theta, \quad i = 1, 2, 3,\ldots, n-1, \tag{22}$$

and

$$r^{-1}\Lambda^s[E-E.c(a,r)] \geqslant \int_0^{2\pi}{}^{*} \Lambda^{s-1}(a;\theta).[E-E.c(a,r)]\,d\theta. \tag{23}$$

Summing each side of (22) over the integers $i$,

$$\sum_{i=1}^{n-1} \rho_i^{-1} \Lambda^s [E.c(a, \rho_{i+1}) - E.c(a, \rho_i)] \geqslant \int_0^{2\pi *} \Lambda^{s-1}(a; \theta) [E.c(a, r) - E.c(a, \delta)]\, d\theta.$$

The limit of the left-hand side, as $n \to \infty$, is the familiar Riemann–Stieltjes integral, and so

$$\int_0^{2\pi *} \Lambda^{s-1}(a; \theta).[E.c(a, r) - E.c(a, \delta)]\, d\theta \leqslant \int_{\rho=\delta}^{r} \rho^{-1}\, d\Lambda^s E.c(a, \rho)$$

$$\leqslant r^{-1} \Lambda^s E.c(a, r) + \int_{\delta}^{r} \rho^{-2} \Lambda^s E.c(a, \rho)\, d\rho.$$

The integrand on the left-hand side is a non-increasing function of $\delta$. By making $\delta \to 0$ the inequality becomes

$$\int_0^{2\pi *} \Lambda^{s-1}(a; \theta).E.c(a, r)\, d\theta \leqslant r^{-1} \Lambda^s E.c(a, r) + \int_0^{r} \rho^{-2} \Lambda^s E.c(a, \rho)\, d\rho.$$

Combining this with (23), we obtain the inequality

$$\int_0^{2\pi *} \Lambda^{s-1}(a; \theta).E\, d\theta \leqslant r^{-1} \Lambda^s E + \int_0^{r} \rho^{-2} \Lambda^s E.c(a, \rho)\, d\rho.$$

At almost all points $a$ of $E$, the upper circular density is finite. In addition $s-1$ is positive, and so for any positive value of $r$, say $r = 1$, the integral on the right-hand side is convergent. It follows that for all such points $a$, the integral on the left-hand side is finite, and the lemma follows.

6.5. Lemmas 17 and 19 enable us to assert

THEOREM III. *Almost every line through almost every point of any $s$-set whose dimension is greater than unity intersects it in a set of dimension $s-1$ and finite $(s-1)$-dimensional measure.*

REMARK. It follows from this theorem that the lines of accessibility at almost every point of any $s$-set whose dimension is greater than unity form a set of measure zero. It is interesting, therefore, that Davies (7) has constructed a set of full (inner) plane measure whose lines of accessibility, at every point, form a set containing $2^{\aleph_0}$ lines in every sector.

6.6. The theorem leaves open the question as to whether the intersections are of positive or zero $(s-1)$-dimensional measure.

Any plane set $E$ which is the cartesian product of an $(s-1)$-set on the $x$-axis and a segment on the $y$-axis is an $s$-set. (This follows from the results of Besicovitch and Moran (5) on product sets.) For almost every point $a$ of $E$ and any angle $\theta$, $\Lambda^{s-1}(a; \theta).E > 0$, and so we have already illustrated

the first possibility. In order to illustrate the second possibility we shall construct a new kind of set.

CONSTRUCTION. It will be understood that $s$ is any fixed number such that $0 < s < 2$.

Given any square $S$ we shall denote the length of its sides by $[S]$. Also, for any positive integer $n$, we define the operation of order $n$ to be performed upon $S$ as follows: Divide each side of $S$ into $n+1$ equal parts by means of $n$ points of subdivision, and so form a lattice of $n^2$ points inside $S$. At each point $a$ of the lattice draw a closed square whose centre is at $a$ and whose sides are of length $n^{-2/s}[S]$ and are inclined at an angle $(n \log n)^{-1}$ to the corresponding sides of $S$ (rotated anti-clockwise).

Now let $S_1$ denote the closed square whose vertices are at the points $(0,0)$, $(1,0)$, $(1,1)$, and $(0,1)$. Perform the operation of order 2 on $S_1$, and denote the union of the squares so formed by $S_2$. Perform the operation of order 3 on each of these squares, and denote the union of all the squares so formed by $S_3$. Repeat the process indefinitely, and so define the closed set

$$F^{(s)} = S_1 S_2 S_3 \ldots .$$

Except for the rotation of the squares in the definition of the operation of order $n$, the method of defining sets of this type is well known. We therefore state without proof that $F^{(s)}$ is an $s$-set.

LEMMA 20. *Irrespective of the choice of inclination of new coordinate axes, the set $F^{(s)}$ so defined satisfies*

$$\int_{-\infty}^{\infty} \Lambda^{s-1} F_\xi^{(s)} \, d\xi = 0, \quad \textit{provided that } 1 < s < 2. \tag{24}$$

*Proof.* For convenience, we shall prove that (24) holds for the coordinate axes used in the definition of $F^{(s)}$. The general case follows by exactly the same method.

For any integer $n$ such that $n > 1$, two sides of each of the squares of the set $S_n$ are inclined at an angle

$$\theta_n = \sum_{u=2}^{n} (u \log u)^{-1} \quad \text{to the } y\text{-axis.}$$

Since this is a divergent series, we can find arbitrarily large values of $n$ such that the principal value of $\theta_n$ differs from zero by at most $(n \log n)^{-1}$.

Suppose that $n$ is any such integer, and denote the squares of the set $S_n$ by

$$S_{n,i}, \quad i = 1, 2, 3, \ldots, m.$$

Consider the operation of order $n+1$ which is performed on any one square $S_{n,i}$. The centres of the squares so formed lie on a set of $n+1$

segments, each of length $[S_{n,i}]$ and inclined at an angle at most $(n\log n)^{-1}$ to the $y$-axis. Let $l$ be any one of these segments. Then

$$\text{proj}_{\frac{1}{2}\pi} l \leqslant [S_{n,i}]\sin(n\log n)^{-1} < (n\log n)^{-1}[S_{n,i}].$$

Denote by $K$ the set of the above squares whose centres lie on $l$. The diameter of each one is equal to $2^{\frac{1}{2}}(n+1)^{-2/s}[S_{n,i}]$ and so

$$\text{diam}(\text{proj}_{\frac{1}{2}\pi} K) < [(n\log n)^{-1}+2^{\frac{1}{2}}(n+1)^{-2/s}][S_{n,i}].$$

Now $$(n\log n)^{-1}+2^{\frac{1}{2}}(n+1)^{-2/s} = o(n^{-1}) \quad \text{as } n\to\infty,$$

and hence we can, given $\epsilon > 0$, assume that we have chosen a sufficiently large (suitable) value of $n$ to ensure that

$$\text{diam}(\text{proj}_{\frac{1}{2}\pi} K) < (n+1)^{-1}\epsilon[S_{n,i}].$$

This means that $K$ can be enclosed in a rectangle $T$ whose base is of length

$$\delta = (n+1)^{-1}\epsilon[S_{n,i}], \tag{25}$$

and whose sides are of length

$$d = [S_{n,i}] \tag{26}$$

and parallel to the $y$-axis.

There being $n+1$ such columns of squares inside each square $S_{n,i}$, $i = 1, 2, 3,..., m$, it follows that the set $F^{(s)}$ can be enclosed in a set of $m(n+1)$ such rectangles $T_j$, $j = 1, 2, 3,..., p$, say, where $p = m(n+1)$. This set of rectangles will satisfy

$$\begin{aligned}\int_{-\infty}^{\infty} \sum_{j=1}^{p} |(T_j)_x|^{s-1}\,dx = \sum_{j=1}^{p} \int_{-\infty}^{\infty} |(T_j)_x|^{s-1}\,dx &= m(n+1)\,\delta d^{s-1} \\ &= m\epsilon[S_{n,i}]^s, \quad \text{from (25) and (26)}, \\ &= \epsilon[S_1]^s, \quad \text{by induction on } n, \\ &= \epsilon.\end{aligned}$$

Setting $[S_{n,i}] = q$, it follows that

$$\int_{-\infty}^{\infty} \Lambda_q^{s-1}(F^{(s)})_x\,dx \leqslant \epsilon.$$

Now $q$ and $\epsilon$ can be taken arbitrarily small by choosing a sufficiently large (suitable) integer $n$, and so the proof of the lemma is complete.

THEOREM IV. *For any value of $s$ such that $1 < s < 2$ there exists an $s$-set $E$ with the following property: almost every line through every point of $E$ intersects $E$ in a set of zero $(s-1)$-dimensional measure.*

*Proof.* Suppose that $s$ is any such number, and consider the set $F^{(s)}$. For every angle $\theta$, denote by $E_\theta$ the set of points $(x, y)$ of $F^{(s)}$ such that

$$\Lambda^{s-1}(x, y; \theta).F^{(s)} > 0.$$

By Lemma 20, $$|\text{proj}_\theta E_\theta| = 0.$$

On the other hand, by Lemma 13. for almost all angles $\theta$, all $s$-sets $E$ which are contained in $F^{(s)}$ satisfy

$$|\mathrm{proj}_\theta E| > 0.$$

It follows that, for almost all angles $\theta$,

$$\Lambda^s E_\theta = 0.$$

Consequently, by the Fubini lemma, almost all points of $F^{(s)}$ satisfy the condition required by this theorem. By removing from $F^{(s)}$ all those points which do not satisfy this condition, we form a set which has the required property, and the proof of the theorem is complete.

## 7. Condensation points and evaporation points of plane sets

7.1. DEFINITIONS. We shall call $a$ a *condensation* point of a plane set $E$ if, for almost every value of $\theta$, $a$ is a limit point of the set $(a;\theta).E$.

If, on the other hand, the set of all such angles $\theta$ is a set of measure zero, then we shall call $a$ an *evaporation* point of $E$.

7.2. THEOREM V. *Almost all points of any s-set whose dimension is greater than unity are condensation points.*

*Proof.* Suppose that we are given an $s$-set $E$, where $s > 1$, and that $r$ is an arbitrary positive number. By dividing the plane into a net of equal squares whose sides are of length $2^{-\frac{1}{2}}r$, and applying Theorem III to the part of $E$ contained in each of them in turn, we see that at almost every point $a$ of $E$, for almost every value of $\theta$ the segment $(a;\theta).c(a,r)$ contains points of $E$. The theorem now follows by making $r \to 0$.

7.3. It is obvious that all points in the plane are evaporation points of any set whose dimension is less than unity. The interesting case is that of linearly measurable sets. A rectifiable curve of finite length has an exact tangent at almost every point, and consequently almost all points of any subset $E$ of such a curve are evaporation points of $E$. It is not very difficult to extend this result to any regular set $E$, using the theorem of Besicovitch ((2), 324) concerning regular sets. On the other hand, the question as to whether the result can be extended to any linearly measurable set remains open. Gillis (10) has shown by means of an example that we can construct irregular sets which project into sets of measure zero from every point in the plane, but sets of this type are of rather a special kind.

## 8. Projections of plane sets from points in the plane

8.1. We have already seen that any $s$-set whose dimension is greater than unity projects into a set of measure $2\pi$ from almost all its points. By subjecting the plane to suitable projective transformations, it is not very difficult to deduce from Theorem I that such a set, in addition, projects

into a set of positive measure from all points in the plane with the exception of a set of Lebesgue plane measure zero. This result also holds for regular linearly measurable sets. On the other hand, by projective transformations, we can deduce from the theorem of Besicovitch (3) that any irregular linearly measurable set projects into a set of measure zero from all points in the plane with the exception of a set of Lebesgue plane measure zero. Professor A. S. Besicovitch suggested that it might be possible to prove the more precise result that the dimension of the exceptional set is at most unity, and it is the purpose of the next paragraph to show that this is so. (The proof depends strongly upon his own theorem.) The question as to whether the linear measure of the exceptional set is necessarily zero (or even finite) is an interesting one, and remains open. We shall see later, by means of an example, that its dimension can be unity.

8.2. LEMMA 21. *The set of values of $x$, $y$ and $\theta$ such that $(x, y; \theta)$ intersects any given closed set $F$ forms a Borel set in $R^3$.*

*Proof.* For every positive integer $n$, the set of values of $x$, $y$ and $\theta$ such that $(x, y; \theta)$ intersects the closure of the set

$$F.c(x, y; n) - F.c(x, y; n^{-1})$$

forms a closed set $F_n$ in $R^3$. The lemma follows by summing the sets $F_n$ over the integers $n$.

NOTATION. By $\text{proj}_{x,y} E$ we shall denote the projection of the plane set $E$ from the point $(x, y)$, i.e. the set of principal values of the angles $\theta$ such that $(x, y; \theta)$ intersects $E$.

LEMMA 22. *If $F$ is a closed set, then $|\text{proj}_{x,y} F|$ is a Borel measurable function of the variables $x$ and $y$.*

*Proof.* For every positive integer $m$, denote by $F_m$ the set of points of $F$ whose distance from the origin does not exceed $m$. Then for any point $(x, y)$ in the plane,

$$\text{proj}_{x,y} F = \text{proj}_{x,y} \sum_{m=1}^{\infty} F_m = \sum_{m=1}^{\infty} \text{proj}_{x,y} F_m,$$

whence

$$|\text{proj}_{x,y} F| = \lim_{m \to \infty} |\text{proj}_{x,y} F_m|. \tag{1}$$

Suppose that $F'$ is any closed bounded set. It is evident from (1) that in order to prove the lemma it is sufficient to prove that the function $|\text{proj}_{x,y} F'|$ is Borel measurable.

For every integer $n$, divide the plane into a net of equal closed squares by lines parallel to the axes, and distance $2^{-n}$ apart. It is understood that the axes themselves are among this set of lines. Denote the set of the squares by

$$S_n = \sum_{i=1}^{\infty} S_{n,i}.$$

Denote the finite subset of the squares which intersect $F'$ by

$$S'_n = \sum_{j=1}^{p(n)} S'_{n,j}.$$

For every pair of positive integers $N, i$, denote by $S^*_{N,i}$ the open square formed by $S_{N,i}$ and the eight adjacent squares of the set $S_N$. Then for every point $(x, y) \in S_{N,i}$, we have

$$\begin{aligned}\text{proj}_{x,y} F'CS^*_{N,i} &= \text{proj}_{x,y} \prod_{n=N}^{\infty} \sum_{j=1}^{p(n)} S'_{n,j} CS^*_{N,i} \\ &= \prod_{n=N}^{\infty} \text{proj}_{x,y} \sum_{j=1}^{p(n)} S'_{n,j} CS^*_{N,i},\end{aligned}$$

whence $$|\text{proj}_{x,y} F'CS^*_{N,i}| = \lim_{n\to\infty} \left|\text{proj}_{x,y} \sum_{j=1}^{p(n)} S'_{n,j} CS^*_{N,i}\right|.$$

The right-hand side of this equality is the limit of a sequence of continuous functions of $x$ and $y$, and so the Borel measurability of the left-hand side, over the set $S_{N,i}$, follows. Consequently, if we let $i(x, y, N)$ denote the integer $i$ such that

$$(x, y) \in S_{N,i} \quad \text{(or the least integer if there are more than one)},$$

then we can define the sequence of Borel measurable functions

$$f_N(x, y) = |\text{proj}_{x,y} F'CS^*_{N,i(x,y,N)}|, \quad N = 1, 2, 3, \ldots.$$

Now

$$\begin{aligned}\text{proj}_{x,y} F' &= \text{proj}_{x,y} \sum_{N=1}^{\infty} F'CS^*_{N,i(x,y,N)} \\ &= \sum_{N=1}^{\infty} \text{proj}_{x,y} F'CS^*_{N,i(x,y,N)},\end{aligned}$$

whence $$|\text{proj}_{x,y} F'| = \lim_{N\to\infty} f_N(x, y).$$

Therefore the left-hand side is a Borel measurable function, and this completes the proof of the lemma.

THEOREM VI. *Every irregular linearly measurable set projects into a set of measure zero from all points in the plane with the exception of a set whose dimension is at most unity.*

*Proof.* Suppose that the theorem is false for a certain irregular set $E$, and express $E$ in the form

$$E = \sum_{i=1}^{\infty} F_i + H,$$

where $\Lambda H = 0$, and $F_i$, $i = 1, 2, 3, \ldots$, are all closed sets. Then the theorem is false for one of these closed sets, say $F$. Let $B$ denote the set of points $(x, y)$ in the plane such that

$$|\text{proj}_{x,y} F| > 0,$$

and let $s$ be one of the numbers greater than unity such that

$$\Lambda^s B = \infty.$$

By Lemma 22, $B$ is a Borel set and consequently, by the theorem of Davies (**6**), we can find a closed $s$-set $F^* \subset B$.

For every angle $\theta$ denote by $E_\theta$ the set of points $(x, y)$ of $F^*$ such that $(x, y; \theta)$ intersects $F$. Denote by $f(x, y, \theta)$ the characteristic function of the set of values of $x$, $y$ and $\theta$ such that $(x, y) \in E_\theta$. Then by Lemma 21 and the Fubini lemma,

$$\int_0^{2\pi} \Lambda^s E_\theta \, d\theta = \int_0^{2\pi} d\theta \int_{F^*} f(x, y, \theta) \, d\Lambda^s(x, y) = \int_{F^*} |\mathrm{proj}_{x,y} F| \, d\Lambda^s(x, y) > 0,$$

that is to say, for a set of positive measure of angles $\theta$,

$$\Lambda^s E_\theta > 0.$$

By Lemma 13, for almost all these angles we have

$$|\mathrm{proj}_\theta E_\theta| > 0.$$

For every angle $\theta$, $\mathrm{proj}_\theta F \supset \mathrm{proj}_\theta E_\theta$, and it follows that $F$ projects into a set of positive measure in a set of directions of positive measure. This contradicts the theorem of Besicovitch (**3**), and so our original assumption is impossible.

## 9. Irregularities in the projections of plane sets

9.1. An irregular linearly measurable set has been constructed by Gillis (**11**) which projects into a set of positive Lebesgue measure in $2^{\aleph_0}$ directions in every sector. We shall modify his example and show that this set of directions can be made, in addition, a set of dimension unity. We have already seen that the theorem of Besicovitch (**3**) is a best possible result in two respects, and our construction will illustrate a third.

First of all, we construct a closed set $F$ on the $x$-axis, dependent upon a (rapidly increasing) sequence of positive integers $m(n)$, $n = 1, 2, 3, \ldots$.

CONSTRUCTION. Let $n$ be any positive integer. Divide the interval $[0, 1]$ into $m(n)$ equal parts, and let each point of subdivision $x = 0, [m(n)]^{-1}, 2[m(n)]^{-1}, \ldots, 1$, be the mid-point of a closed interval of length $n^{-2}[m(n)]^{-1}$. Denote this set of $m(n)+1$ intervals by $F_n$. Having defined such a set $F_n$ for every positive integer $n$, we then define the set

$$F = F(m(1), m(2), m(3), \ldots) = \prod_{n=1}^{\infty} F_n.$$

LEMMA 23. *Provided that the sequence $m(n)$, $n = 1, 2, 3, \ldots$, is suitably chosen the dimension of the corresponding set $F$ is equal to unity.*

*Proof.* We shall define a suitable sequence $m(n)$, $n = 1, 2, 3, \ldots$, inductively, having assigned to $m(1)$ the value unity.

Suppose that we have defined $m(u)$, $u = 1, 2, 3, \ldots, n$, and hence that we have determined the corresponding sets $F_u$, $u = 1, 2, 3, \ldots, n$. Now whatever

value we assign to $m(n+1)$, the ratio of the length of each interval of the set $F_{n+1}$ to the distance apart of the mid-points of any two adjacent ones is equal to $(n+1)^{-2}$. This number, however small it may be, is quite independent of our choice of $m(n+1)$. Consequently, if we denote by

$$\sum_{j=1}^{p} I_j$$

the set of all those intervals of $F_{n+1}$ which lie inside the determined set of intervals

$$\prod_{u=1}^{n} F_u,$$

and let

$$t(n) = 1-n^{-1}, \tag{1}$$

then by making $m(n+1)$ sufficiently large, we can make as large as we please the quantity

$$\sum_{j=1}^{p} |I_j|^{t(n)}.$$

We choose a value of $m(n+1)$ for which this quantity is greater than unity, and then proceed to define $m(n+2)$ in a similar way. The required sequence $m(n)$, $n = 1, 2, 3,...$, is formed by repeating the process indefinitely. Thus we have determined the set $F$, and for every positive integer $n$,

$$\Lambda^{t(n)}F \geqslant 1,$$

and so from (1) the lemma follows.

We now pass on to our main construction.

CONSTRUCTION. Let $m(n)$, $n = 1, 2, 3,...$, denote a sequence of positive integers which satisfies the condition of Lemma 23. Suppose that $R$ is any closed rectangle whose base is parallel to the $x$-axis, and whose sides are of a length which divides into the length of the base an integral number of times, and suppose that $n$ is any positive integer. We shall now define the operation of order $n$ which is to be performed upon $R$. For simplicity of notation, we shall *temporarily* assume that the vertices of $R$ are at the points $A(0,0)$, $A'(w,0)$, $B'(w,1)$, and $B(0,1)$, where $w$ is an integer. Let $A_i$, $i = 1, 2, 3,..., 2wm(n)$ denote the points $(\frac{1}{2}i[m(n)]^{-1}, 0)$ on $AA'$, and let $B_i$ denote the points $(\frac{1}{2}i[m(n)^{-1}], 1)$ on $BB'$. Upon the segments

$$AA_1,\ B_1B_2,\ A_2A_3,\ B_3B_4,...,\ B_{2wm(n)-1}B_{2wm(n)}$$

draw closed rectangles as shown in Fig. 2. The length of the vertical sides of each rectangle is taken as $\frac{1}{2}n^{-2}[m(n)]^{-1}$. Extend each end of every rectangle by means of two closed squares. This set of $2wm(n)$ extended rectangles constitutes the rectangles *formed by the operation.*

Now to construct our set, let $T_1$ denote the rectangle whose vertices are at the points $(0,0)$, $(3,0)$, $(3,1)$, and $(0,1)$. Perform the operation of order 2

upon $T_1$, and denote the set of rectangles so formed by $T_2$. Perform the operation of order 3 upon every rectangle of $T_2$, and denote the set of all the rectangles so formed by $T_3$. Having repeated the process indefinitely, we define the set

$$T = \prod_{n=1}^{\infty} T_n.$$

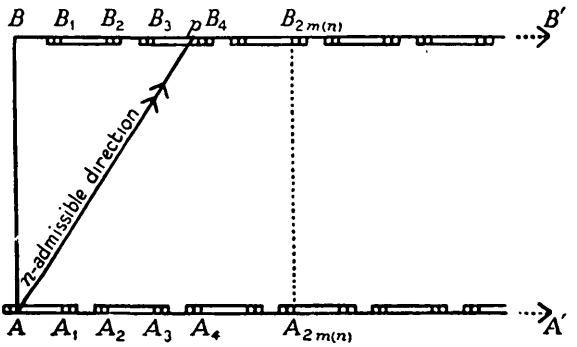



FIG. 2

LEMMA 24. *The set of directions $\theta$ such that $|\mathrm{proj}_\theta T| > 0$ is of dimension unity.*

For the purpose of the proof, we shall call $\theta$ an *admissible* direction if every line which lies in the direction $\theta$ and which intersects both of the horizontal sides of $T_1$, intersects the set $T$.

We shall call $\theta$ an *n-admissible* direction if every line which lies in the direction $\theta$ and which intersects both of the horizontal sides of some rectangle $R$ of the set $T_n$, intersects both of the horizontal sides of some rectangle formed by the operation of order $n+1$ which was performed upon $R$.

*Proof.* Let $E$ denote the set of cotangents of all admissible directions, and $E_n$ the set of cotangents of all $n$-admissible directions. Thus

$$E \supset \prod_{n=1}^{\infty} E_n$$

and $\quad |\mathrm{proj}_\theta T| \geqslant 2^{-\frac{1}{2}}$ whenever $\cot\theta \in E$ and $\frac{1}{4}\pi \leqslant \theta \leqslant \frac{1}{2}\pi$.

Hence, in order to prove the lemma, it is sufficient to prove that the set

$$[0,1].\prod_{n=1}^{\infty} E_n \tag{2}$$

is of dimension unity. To this end we investigate the structure of the sets $E_n$, $n = 1, 2, 3, \ldots$.

Suppose that $n$ is any positive integer, and that the diagram represents some rectangle $R$ of the set $T_n$. For every even integer $i$ such that

$$0 \leqslant i \leqslant 2m(n),$$

draw a closed interval $J_i$ parallel to the $x$-axis, of length $n^{-2}[m(n)]^{-1}$, and with $B_i$ as mid-point. Then for every point $p$ which belongs to one of these intervals, the direction of $Ap$ is $n$-admissible (see Fig. 2, where $p$ is taken near $B_4$). Consequently the set $E_n$ contains the set $F_n$ which was defined in the process of constructing the linear set $F$. It follows that the set (2) is of dimension unity, and the lemma is established.

LEMMA 25. *The set $T$ is an irregular linearly measurable set.*

*Proof.* By Lemma 24 it is obvious that

$$\Lambda T > 0. \tag{3}$$

For every positive integer $n$ denote by $l_n$ the sum of the lengths of the bases of the rectangles of the set $T_n$, and by $q(n)$ the diameter of each of these rectangles. Then we have

$$l_{n+1} = [1+4(n+1)^{-2}]l_n$$

and

$$\Lambda_{q(n)}T < 2l_n, \quad n = 1, 2, 3,\ldots,$$

whence

$$\lim_{n\to\infty} \Lambda_{q(n)}T < \infty.$$

Since

$$\lim_{n\to\infty} q(n) = 0,$$

it follows that

$$\Lambda T < \infty, \tag{4}$$

and so from (3), the set $T$ is linearly measurable.

Finally, at almost every point of $T$ there is no tangent, and hence the set is also irregular.

Lemmas 24 and 25 enable us to assert

THEOREM VII. *There exists an irregular linearly measurable set which projects into a set of positive Lebesgue measure in a set of directions whose dimension is equal to unity.*

REMARK. We can modify the set $T$ by subjecting the plane to any non-affine projective transformation which does not transform any point of $T$ into a point on the line at infinity. Under any such transformation let $T$ be transformed into a set of points $T'$, and the line at infinity into a line $l$. Then $T'$ is an irregular linearly measurable set, and it projects into a set of positive measure from a set of points (on $l$) whose dimension is equal to unity. This shows us that Theorem VI is the best possible result *dimensionally.*

9.2. Besicovitch and Moran (5) have proved that the dimension of the cartesian product set of any $\alpha$-dimensional set on the $x$-axis and any $\beta$-dimensional set on the $y$-axis is at most $1+\min(\alpha,\beta)$. If in particular we take the set on the $y$-axis to be the entire axis, then its dimension is equal to unity and it follows that the dimension of the corresponding product set is at most $1+\alpha$. Immediately we deduce

LEMMA 26. *If $E$ is an $s$-set, and $s \geqslant 1$, then for every angle $\theta$ the dimension of the set* $\mathrm{proj}_\theta E$ *is at least* $s-1$.

On the other hand, it follows from Theorem I that the set of angles $\theta$ for which this dimensional number is actually equal to $s-1$ is an exceptional set and of measure zero. We shall construct an example to show that it is possible for the exceptional set to contain $2^{\aleph_0}$ angles in every sector. Our construction is based largely on the idea of a set constructed by Besicovitch and Moran (5). First of all we state the following lemma, the proof of which is left to the reader.

LEMMA 27. *Suppose that $(a,b)$ is an interval on the $x$-axis, that $I_n$, $n = 1, 2, 3,\ldots$, is a sequence of intervals on the $x$-axis, and that for every positive integer $n$, the set*

$$E_n = \sum_{u=n}^{\infty} I_u$$

*is everywhere dense in $(a,b)$. Then the set*

$$E = \prod_{n=1}^{\infty} E_n = \overline{\lim_{n\to\infty}} I_n$$

*contains $2^{\aleph_0}$ points in every interval which is contained in $(a,b)$.*

CONSTRUCTION. We shall define an operation to be performed upon any given closed circle $C$, dependent upon any given angle $\theta$ and positive numbers $\epsilon$, $q$, $s$, and $t$ such that $1 < s < 2$ and $t > s-1$. The aim of the operation is to form a sequence of non-overlapping closed circles $C_i$, $i = 1, 2, 3,\ldots, m$, contained in $C$ such that

$$\sum_{i=1}^{m} (dC_i)^s = (dC)^s \tag{5}$$

and

$$\Lambda_q^t \mathrm{proj}_\theta \sum_{i=1}^{m} C_i < \epsilon. \tag{6}$$

In addition, the diameters of all the circles are to be equal to each other, and in a certain sense the circles are to be uniformly scattered throughout $C$. These facts ensure that the set which we shall construct by means of a sequence of such operations upon circles will be an $s$-set, and we shall assume this assertion without proof.

Now to perform the operation, we first choose any positive integer $n$,

and later impose conditions upon its magnitude. Divide the diameter of $C$ whose direction is $\theta+\frac{1}{2}\pi$ into $n+1$ equal parts, and through every point of subdivision draw a chord $l_{n,i}$ in the direction $\theta$. Draw as many as possible non-overlapping open circles of diameter

$$d(n) = (\tfrac{1}{2}n)^{1/(1-s)}\, dC \tag{7}$$

inside $C$ such that the centre of every circle lies on some chord $l_{n,i}$, $i = 1, 2, 3,..., n$. Denote the total number of circles so constructed by $m(n)$.

Now regarding $n$ as a variable integer, and $d(n)$ and $m(n)$ as functions of $n$, we have

$$\sum_{i=1}^{n} \Lambda l_{n,i} \sim m(n)\, d(n) \quad \text{as } n \to \infty. \tag{8}$$

Quite independently of $m(n)$ and $d(n)$ we have

$$|C| = \lim_{n\to\infty} n^{-1}\, dC \sum_{i=1}^{n} \Lambda l_{n,i},$$

whence

$$\tfrac{1}{4}\pi\, dC = \lim_{n\to\infty} n^{-1} \sum_{i=1}^{n} \Lambda l_{n,i} = \lim_{n\to\infty} n^{-1}\, m(n)\, d(n), \quad \text{from (8).}$$

Now from (7), $\qquad n^{-1} = \tfrac{1}{2}(dC)^{1-s}\, d(n)^{s-1},$

and so $\qquad \lim_{n\to\infty} m(n)\, d(n)^s = \tfrac{1}{2}\pi (dC)^s.$

Therefore for all sufficiently large values of $n$,

$$m(n)\, d(n)^s > (dC)^s,$$

and in this case we can reduce the radii of every one of the $m(n)$ circles by equal amounts and so form a new sequence of circles $C_{n,i}$, $i = 1, 2, 3,..., m(n)$, which satisfies (5). It remains to show that for sufficiently large values of $n$, (6) is also satisfied.

Now the set $\qquad \operatorname{proj}_\theta \sum_{i=1}^{m(n)} C_{n,i}$

can be enclosed in a set of $n$ intervals each of length less than $d(n)$, and hence

$$\Lambda^t_{d(n)} \operatorname{proj}_\theta \sum_{i=1}^{m(n)} C_{n,i} < n\, d(n)^t. \tag{9}$$

Since $t > s-1$ and $s > 1$, it follows from (7) that

$$\lim_{n\to\infty} n\, d(n)^t = 0,$$

and in particular that $\qquad \lim_{n\to\infty} d(n) = 0.$

Consequently the inequality (6) may be deduced from (9) provided that $n$ is sufficiently large. It is therefore possible to perform an operation upon $C$ of the required kind.

We now pass on to the construction of a closed set $F_1^{(s)}$, given a number $s$ such that $1 < s < 2$. We define the following functions of an integer $p$:

$$\theta(p) = \sum_{j=1}^{p} j^{-1}, \tag{10}$$

$$q(p) = p^{-1} \quad \text{and} \quad t(p) = p^{-1} + s - 1. \tag{11}$$

We shall define the function $\epsilon(p)$ by an inductive process.

By the *operation of order* $p$ upon a circle $C$ we mean the operation we have already defined, dependent upon $\theta(p)$, $\epsilon(p)$, $q(p)$, $s$, and $t(p)$. Let $C^{(1)}$ denote any closed circle, and let $\epsilon(1)$ be any positive number, say unity. Suppose that $\epsilon(j), j = 1, 2, 3, \ldots, p-1$, have been defined and that we wish to define $\epsilon(p)$. Perform the operation of order 2 upon $C^{(1)}$, forming a set of circles $C^{(2)}$. Upon each one of these circles perform the operation of order 3, and denote the set of all the circles so formed by $C^{(3)}$. Repeat this process $p-2$ times, and let $w(p)$ denote the number of circles of the set $C^{(p-1)}$. We define

$$\epsilon(p) = p^{-1}[w(p)]^{-1}.$$

From (6) this ensures that if we perform the operation of order $p$ upon every one of these $w(p)$ circles, and denote the set of all the circles so formed by $C^{(p)}$, we shall have

$$\Lambda_{q(p)}^{t(p)} \operatorname{proj}_{\theta(p)} C^{(p)} < p^{-1},$$

and hence, by continuity arguments, there will exist a positive number $\delta(p)$ such that

$$\Lambda_{q(p)}^{t(p)} \operatorname{proj}_{\theta} C^{(p)} < p^{-1} \quad \text{whenever } |\theta - \theta(p)| < \delta(p). \tag{12}$$

In the process of inductively defining $\epsilon(p)$, $p = 1, 2, 3, \ldots$, we form the sequence of sets $C^{(p)}$. We now define the set

$$F_1^{(s)} = \prod_{p=1}^{\infty} C^{(p)}.$$

From (10) the principal values of the angles $\theta(p)$, $p = 1, 2, 3, \ldots$, form an everywhere dense set in the interval $(0, 2\pi)$, and it follows from Lemma 27 that the set $\Psi$ of angles $\theta$ which satisfy (12) for infinitely many integers $p$, contains $2^{\aleph_0}$ angles in every sector. Thus we have

$$\lim_{p \to \infty} \Lambda_{q(p)}^{t(p)} \operatorname{proj}_{\theta} F_1^{(s)} = 0 \quad \text{whenever } \theta \in \Psi.$$

From (11) $\quad \lim_{p \to \infty} q(p) = 0 \quad \text{and} \quad \lim_{p \to \infty} t(p) = s - 1,$

and consequently

$$\Lambda^t \operatorname{proj}_{\theta} F_1^{(s)} = 0 \quad \text{whenever } \theta \in \Psi \text{ and } t > s-1.$$

That is to say, the dimension of the set

$$\operatorname{proj}_{\theta} F_1^{(s)}$$

is at most $s-1$ for all such angles $\theta$.

By virtue of Lemma 26 we are now able to assert

THEOREM VIII. *For every value of $s$ such that $1 < s < 2$, there exists an $s$-set which projects into a set of dimension $s-1$ in $2^{\aleph_0}$ angles in every sector.*

REMARK. We can modify the construction of $F_1^{(s)}$ in various ways. The number $s$, which has previously been fixed, can be a function $s(p)$ of the integer $p$, but we must have $t(p) > s(p)-1$ and $1 < s(p) < 2$. For example, if we set

$$s(p) = 1+\tfrac{1}{2}p^{-1}, \qquad t(p) = p^{-1},$$

then the corresponding set, $F_1^{(1)}$ say, is of infinite linear measure and it projects into a set of dimension zero in an exceptional set of directions. On the other hand, if we set

$$s(p) = 2-p^{-1}, \qquad t(p) = 1,$$

then the corresponding set, $F_1^{(2)}$ say, is of dimension two and it projects into a set of Lebesgue measure zero in an exceptional set of directions.

## CHAPTER III

## 10. Angular density of plane $s$-sets

10.1. NOTATION AND DEFINITIONS. Given a point $a$ and a pair of angles $\theta_1$ and $\theta_2$ such that $\theta_1 < \theta_2$, by $(a;\theta_1,\theta_2)$ we denote the set of points which lie on any ray $(a;\theta)$ where $\theta_1 \leqslant \theta \leqslant \theta_2$. Suppose, in addition, that $r$ is a positive number. Then by $(a;\theta_1,\theta_2,r)$ we denote the set $(a;\theta_1,\theta_2).c(a,r)$, that is to say, a certain sector of the circle $c(a,r)$.

We shall define the *upper and lower angular densities* of a plane $s$-set $E$ at $a$ in the *angle* $(\theta_1,\theta_2)$ as the upper and lower limits, as $r \to 0$, of the ratio

$$\frac{\Lambda^s E.(a;\theta_1,\theta_2,r)}{2^s r^s}.$$

In the particular case $s = 1$ we have the same definition of angular density as that given by Besicovitch ((2), 297) for linearly measurable sets. Besicovitch (4) has investigated the one-sided density of $s$-sets which lie on a straight line. In certain respects we may regard angular density of $s$-sets in the plane as a generalization of this idea.

10.2. Besicovitch ((2), 307–10) has shown that at almost all points of any irregular linearly measurable set, the sum of the upper angular densities in any pair of opposite angles $(\theta_1,\theta_2)$ and $(\theta_1+\pi,\theta_2+\pi)$ is positive, and greater than $\frac{1}{3}\sin\frac{1}{2}(\theta_2-\theta_1)$. His proof depends upon the fact that no rectifiable curve intersects such a set in a set of positive linear measure.

Now it is obvious that no rectifiable curve intersects any $s$-set whose dimension exceeds unity in a set of positive $s$-dimensional measure, and for this reason the above theorem also holds for sets of this class. We shall

prove that, more precisely, at almost all points of such a set the angular density is positive in every single angle $(\theta_1, \theta_2)$. On the other hand, Besicovitch ((2), 327–8) has given an example to show that this more precise theorem is not true for irregular linearly measurable sets. Dickinson (8) has modified his example and gone into greater detail.

NOTATION. By $H(a;\theta,\delta,r)$ we shall denote the set

$$(a;\theta-\tfrac{1}{2}\pi,\theta+\tfrac{1}{2}\pi).S(a;\theta,\delta,r).$$

LEMMA 28. *Suppose that $\phi$ and $\psi$ are any two angles such that*

$$0 \leqslant \phi < \psi \leqslant 2\pi$$

*and that $E$ is an $s$-set. Then at almost all points $(x,y)$ of $E$,*

$$\overline{\lim_{r\to 0}}\,\overline{\lim_{\delta\to 0}}\,\delta^{-1}r^{1-s}\int_{\phi}^{\psi}\Lambda^s E.H(x,y;\theta,\delta,r)\,d\theta > 2^{-8}(\psi-\phi).$$

*Proof.* Suppose that $r$ is an arbitrary positive number. Then in order to prove the lemma, it is sufficient to prove that at almost all points $(x,y)$ of $E$,

$$\sup_{\rho<r}\overline{\lim_{\delta\to 0}}\,\delta^{-1}\rho^{1-s}\int_{\phi}^{\psi}\Lambda^s E.H(x,y;\theta,\delta,2\rho)\,d\theta > 2^{-7}(\psi-\phi). \tag{1}$$

By Lemma 3, at almost all points of $E$ the upper circular density is greater than or equal to $2^{-s}$. Suppose that $a$ is any such point. Then there exists an arbitrarily small positive number $\rho$ such that

$$\rho < r \quad \text{and} \quad \Lambda^s E.c(a,\rho) > \tfrac{1}{2}\rho^s. \tag{2}$$

Having fixed $a$ and $\rho$, choose any angle $\theta$ and any positive integer $n$. Divide the diameter of $c(a,\rho)$ which lies in the direction $\theta+\frac{1}{2}\pi$ into $n$ equal parts, and through each point of subdivision draw a chord in the direction $\theta$. In this way we divide the circle into $n$ sections, $S_u$, $u = 1, 2, 3,..., n$. Let $T_i$, $i = 1, 2, 3,..., m$, denote all those sections for which

$$\Lambda^s E T_i > \tfrac{1}{2}n^{-1}\Lambda^s E.c(a,\rho), \tag{3}$$

and let $T'_j$, $j = 1, 2, 3,..., p$, denote all the others. Then from (3)

$$\sum_{j=1}^{p}\Lambda^s E T'_j \leqslant \tfrac{1}{2}pn^{-1}\Lambda^s E.c(a,\rho) \leqslant \tfrac{1}{2}\Lambda^s E.c(a,\rho),$$

and consequently

$$\sum_{i=1}^{m}\Lambda^s E T_i \geqslant \tfrac{1}{2}\Lambda^s E.c(a,\rho). \tag{4}$$

Now each section $T_i$, $i = 1, 2, 3,..., m$, can be divided into two parts $T_i^{(1)}$ and $T_i^{(2)}$ by a segment in the direction $\theta+\frac{1}{2}\pi$ in such a way that

$$\Lambda^s E T_i^{(1)} = \Lambda^s E T_i^{(2)}.$$

By virtue of (4) and (3) respectively, it is seen that in this way we can find a set $E_\theta \subset E.c(a,\rho)$ such that

$$\Lambda^s E_\theta \geqslant \tfrac{1}{4}\Lambda^s E.c(a,\rho), \tag{5}$$

and such that for every point $(x,y)$ of $E_\theta$,

$$\Lambda^s E.H(x,y;\theta,2n^{-1}\rho,2\rho) > \tfrac{1}{4}n^{-1}\Lambda^s E.c(a,\rho). \tag{6}$$

We now define $f(x,y,\theta)$ as the characteristic function of the set of values of $x$, $y$ and $\theta$ such that (6) holds. By the Fubini lemma and from (5),

$$\int_{E.c(a,\rho)} d\Lambda^s(x,y) \int_\phi^\psi f(x,y,\theta)\,d\theta \geqslant \int_\phi^\psi \Lambda^s E_\theta\, d\theta \geqslant \tfrac{1}{4}(\psi-\phi)\Lambda^s E.c(a,\rho).$$

On the other hand, for every point $(x,y)$ it is obvious that

$$\int_\phi^\psi f(x,y,\theta)\,d\theta \leqslant \psi-\phi,$$

and so it follows that there exists a set $E_n \subset E.c(a,\rho)$ such that

$$\Lambda^s E_n \geqslant \tfrac{1}{8}\Lambda^s E.c(a,\rho), \tag{7}$$

and such that for every point $(x,y)$ of $E_n$ we have

$$\int_\phi^\psi f(x,y,\theta)\,d\theta > \tfrac{1}{8}(\psi-\phi),$$

and hence from (6)

$$\int_\phi^\psi \Lambda^s E.H(x,y;\theta,2n^{-1}\rho,2\rho)\,d\theta > 2^{-5}n^{-1}(\psi-\phi)\Lambda^s E.c(a,\rho). \tag{8}$$

For every positive integer $n$ we can find such a set $E_n$ and then define the set

$$A = \overline{\lim_{n\to\infty}} E_n \subset c(a,\rho).$$

From (7) and (2), $\qquad \Lambda^s A \geqslant \tfrac{1}{8}\Lambda^s E.c(a,\rho) > 2^{-4}\rho^s,$ (9)

and from (8), for every point $(x,y)$ of $A$, we have

$$\overline{\lim_{n\to\infty}}\, n\int_\phi^\psi \Lambda^s E.H(x,y;\theta,2n^{-1}\rho,2\rho)\,d\theta \geqslant 2^{-5}(\psi-\phi)\Lambda^s E.c(a,\rho),$$

and hence

$$\overline{\lim_{\delta\to 0}}\, \delta^{-1}\rho \int_\phi^\psi \Lambda^s E.H(x,y;\theta,\delta,2\rho)\,d\theta \geqslant 2^{-6}(\psi-\phi)\Lambda^s E.c(a,\rho) > 2^{-7}(\psi-\phi)\rho^s,$$

from (2);

that is to say, (1) holds. Thus from (9) the set of points of $E$ at which (1) holds has a mean density greater than $2^{-4-s}$ in the circle $c(a,\rho)$. Since there exist arbitrarily small values of $\rho$ which satisfy (2), we deduce that

the upper circular density at $a$ of this set of points is not less than $2^{-4-s}$ Consequently, at almost all points of $E$ the upper circular density of the set of points at which (1) holds is positive, and so by using Lemma 4 we are able to complete our proof of Lemma 28.

THEOREM IX. *At almost all points of any s-set the upper angular density in every angle $(\theta_1, \theta_2)$ exceeds $(s-1)(\theta_2-\theta_1)K$, where $K$ is a positive absolute constant.*

*Proof.* We may suppose, for an obvious reason, that $s > 1$.

Suppose that $\theta_1$ and $\theta_2$ are any two angles such that $0 \leqslant \theta_1 < \theta_2 \leqslant 2\pi$, $\theta_2-\theta_1 \leqslant \pi$, and that $a$ is any point of $E$ at which the upper circular density does not exceed unity. Let $\lambda = \lambda(a)$ be the upper angular density at $a$ in

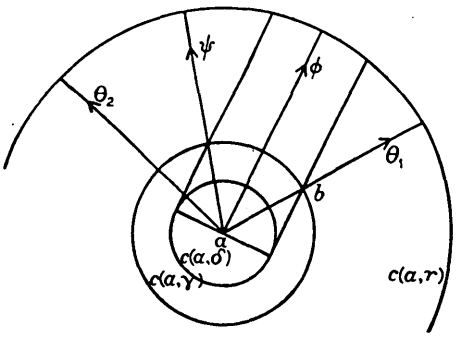



FIG. 3

the angle $(\theta_1, \theta_2)$. Then we can, given $\epsilon > 0$, choose an arbitrarily small positive number $r$ such that whenever $\rho \leqslant r$,

$$\Lambda^s E.(a; \theta_1, \theta_2, \rho) < 2^{s+1}(\lambda+\epsilon)\rho^s \tag{10}$$

and

$$\Lambda^s E.c(a, \rho) < 2^{s+1}\rho^s. \tag{11}$$

Suppose that $\delta$ is an arbitrary positive number less than $r$. Set

$$\theta_1+\tfrac{1}{3}(\theta_2-\theta_1) = \phi \quad \text{and} \quad \theta_2-\tfrac{1}{3}(\theta_2-\theta_1) = \psi, \tag{12}$$

and let $\gamma = \gamma(\delta)$ be the smallest positive number $\rho$ such that for every angle $\theta$ in the range $(\phi, \psi)$ we shall have

$$H(a; \theta, \delta, r) \subset (a; \theta_1, \theta_2, r)+c(a, \rho). \tag{13}$$

Clearly, $\delta < \gamma \leqslant r$.

Now it follows from the way we have defined $\gamma$ that, provided

$$\delta \leqslant r\sin(\phi-\theta_1),$$

the circle $c(a, \gamma)$ passes through the point of intersection $b$ of the right-hand side of $H(a; \phi, \delta, r)$ with the line $(a; \theta_1)$ (as in Fig. 3). If, on the other hand, $\delta > r\sin(\phi-\theta_1)$, then $\gamma = r$. By elementary geometry,

$$\frac{\delta}{\gamma} \geqslant \sin(\phi-\theta_1) > \tfrac{1}{2}(\phi-\theta_1),$$

whence
$$\gamma < \frac{2\delta}{\phi-\theta_1}. \tag{14}$$

From (13), for every angle $\theta$ in the range $(\phi, \psi)$, we have
$$H(a;\theta,\delta,r) \subset (a;\theta_1,\theta_2,r).S(a;\theta,\delta,r)+S(a;\theta,\delta,\gamma),$$
and hence
$$\int_\phi^\psi \Lambda^s E.H(a;\theta,\delta,r)\,d\theta \leqslant \int_0^{2\pi} \Lambda^s E.(a;\theta_1,\theta_2,r).S(a;\theta,\delta,r)\,d\theta + \int_0^{2\pi} \Lambda^s E.S(a;\theta,\delta,\gamma)\,d\theta.$$

We apply Lemma 8 to each of the integrals on the right-hand side, and then use (10) and (11). This gives
$$\begin{aligned}\delta^{-1}\int_\phi^\psi \Lambda^s E.H(a;\theta,\delta,r)\,d\theta &< 2\pi\Big[r^{-1}\Lambda^s E.(a;\theta_1,\theta_2,r)+\gamma^{-1}\Lambda^s E.c(a,\gamma)+ \\ &\qquad + \int_\delta^r \rho^{-2}\Lambda^s E.(a;\theta_1,\theta_2,\rho)\,d\rho + \int_\delta^\gamma \rho^{-2}\Lambda^s E.c(a,\rho)\,d\rho\Big] \\ &< 2^{s+2}\pi\Big[(\lambda+\epsilon)r^{s-1}+\gamma^{s-1}+\frac{\lambda+\epsilon}{s-1}r^{s-1}+\frac{1}{s-1}\gamma^{s-1}\Big] \\ &< \frac{2^7}{s-1}[(\lambda+\epsilon)r^{s-1}+\gamma^{s-1}].\end{aligned}$$

From (14) the limit, as $\delta \to 0$, of the second term on the right-hand side is equal to zero, and so
$$\overline{\lim_{\delta\to 0}}\,\delta^{-1}r^{1-s}\int_\phi^\psi \Lambda^s E.H(a;\theta,\delta,r)\,d\theta \leqslant \frac{2^7}{s-1}(\lambda+\epsilon),$$
and since this holds for all sufficiently small values of $r$,
$$\overline{\lim_{r\to 0}}\,\overline{\lim_{\delta\to 0}}\,\delta^{-1}r^{1-s}\int_\phi^\psi \Lambda^s E.H(a;\theta,\delta,r)\,d\theta \leqslant \frac{2^7}{s-1}(\lambda+\epsilon). \tag{15}$$

Now $\epsilon$ is arbitrary and so, using Lemma 28, we deduce from (15) that at almost all points $a$ of $E$,
$$\begin{aligned}\lambda = \lambda(a) &\geqslant 2^{-15}(s-1)(\psi-\phi) \\ &> 2^{-17}(s-1)(\theta_2-\theta_1), \quad \text{from (12).}\end{aligned} \tag{16}$$

In order to fix the ideas, we supposed that $\theta_2-\theta_1 \leqslant \pi$, but this was unnecessary. In any case it follows from (16) that in any angle $(\theta_1,\theta_2)$ the upper density exceeds $2^{-18}(s-1)(\theta_2-\theta_1)$ at almost all points of $E$. By

arranging all pairs of rational angles $\theta_1$, $\theta_2$ in a sequence we are able to deduce the theorem as stated, and a suitable value of $K$ is $2^{-18}$.

10.3. Gillis (**12**) has shown that at almost all points of any irregular linearly measurable set, the lower angular density is zero in every angle $(\theta+\eta, \theta+\pi-\eta)$, where $\eta$ is an arbitrarily small positive number. Besicovitch ((**2**), 317–20) later proved that, more precisely, this theorem holds with $\eta = 0$. We now prove

THEOREM X. *At almost every point of any s-set whose dimension lies strictly between zero and unity, the lower angular density is zero in every angle* $(\theta+\eta, \theta+\pi-\eta)$, *where* $\eta$ *is an arbitrary positive number.*

*Proof.* Suppose, contrary to the theorem, that there exists an $s$-set $E$, where $0 < s < 1$, a closed $s$-set $F \subset E$, and positive numbers $\eta$, $r$, and $\delta$ such that for every point $a$ of $F$,

$$\Lambda^s E.(a; \eta, \pi-\eta, \rho) > \delta\rho^s \quad \text{whenever } \rho < r_1. \tag{17}$$

Let $a_1$ be any point of $F$ at which the upper circular density of $E$, say $2^{-s}\mu = 2^{-s}\mu(a_1)$, is finite, and at which the circular density of $E-F$ is zero. Then we can, given $\epsilon > 0$, find a positive number $r_2 < r_1$ such that whenever $\rho < r_2$,

$$\Lambda^s F.c(a_1, \rho) \leqslant \Lambda^s E.c(a_1, \rho) < (\mu+\epsilon)\rho^s \tag{18}$$

and

$$\Lambda^s(E-F).c(a_1, \rho) < \epsilon\rho^s. \tag{19}$$

In addition, we can find a positive number $\rho_1$,

$$\rho_1 < \tfrac{1}{2}r_2, \tag{20}$$

such that

$$\Lambda^s E.c(a_1, \rho_1) > (\mu-\epsilon)\rho_1^s,$$

and hence from (19),

$$\Lambda^s F.c(a_1, \rho_1) > (\mu-2\epsilon)\rho_1^s. \tag{21}$$

Let $a_2$ be the point, or one of the points, of the set $F.c(a_1, \rho_1)$ whose ordinate is greatest. From (17), (19), and (20),

$$\Lambda^s F.(a_2; \eta, \pi-\eta, \rho) > \delta\rho^s - \epsilon(\rho_1+\rho)^s \quad \text{whenever } \rho < \rho_1,$$

and together with (21) this gives

$$\Lambda^s F.c(a_1, \rho_1+\rho) > (\mu-2\epsilon)\rho_1^s + \delta\rho^s - \epsilon(\rho_1+\rho)^s.$$

Using (18) and (20), we deduce that whenever $\rho < \rho_1$,

$$(\mu+2\epsilon)(\rho_1+\rho)^s > (\mu-2\epsilon)\rho_1^s + \delta\rho^s,$$

that is to say, for every positive number $\alpha$ less than unity,

$$(\mu+2\epsilon)(1+\alpha)^s > (\mu-2\epsilon)+\delta\alpha^s,$$

and since $\epsilon$ is arbitrary it follows that

$$\mu(1+\alpha)^s \geqslant \mu+\delta\alpha^s. \tag{22}$$

On the other hand,

$$(1+\alpha)^s = 1+O(\alpha) \quad \text{as } \alpha \to 0.$$

This contradicts (22) and hence our original assumption must be false. Therefore the theorem is proved for $\theta = 0$, and the general case follows by the same method.

10.4. The question naturally arises as to whether Theorem X necessarily holds for $s$-sets of dimension greater than unity. However, a simple example shows us that this is not the case.

Suppose that $s$ is a number such that $1 < s < 2$, and that $A$ is an $(s-1)$-set on the $x$-axis, almost all points of which are points of positive lower two-sided density. Define the plane $s$-set $E$ as the cartesian product set of $A$ and any finite segment on the $y$-axis, say $[0, 1]$. It is not difficult to see that at almost all points of $E$ the lower angular density is positive in every angle $(\theta, \theta+\pi)$ provided that $\theta$ is not equal to either $\frac{1}{2}\pi$ or $\frac{3}{2}\pi$. On the other hand, when $\theta = \frac{1}{2}\pi$ or $\theta = \frac{3}{2}\pi$, this density is equal to zero at almost all points of $E$. For by a theorem of Besicovitch ((4), 169–174), the lower one-sided density of $A$ is equal to zero at almost all its points.

In this paragraph we prove a theorem concerning lower angular density which attempts to generalize the idea of this example. Our theorem will have an important application in the next section.

LEMMA 29. *At almost every point $a$ of any $s$-set $E$, where $s > 1$,*

$$\lim_{\delta \to 0} \overline{\lim_{r \to 0}} \frac{\Lambda^s E.(a; \theta-\delta, \theta+\delta, r)}{r^s} = 0 \quad \textit{for all values of } \theta. \tag{23}$$

*Proof.* Suppose that $\epsilon$ is an arbitrary positive number. Then we can find a set $E_1 \subset E$ and a positive number $r_1$ such that

$$\Lambda^s E_1 > \Lambda^s E - \epsilon \tag{24}$$

and such that for every point $a$ of $E_1$,

$$\Lambda^s E.c(a, r) < 2^{s+1} r^s \quad \text{whenever } r < r_1. \tag{25}$$

Let $a_1$ be any point of $E_1$, let $r$ and $\delta$ be arbitrary positive numbers such that $r < r_1$ and $\delta < \frac{1}{8}\pi$, and let $\theta$ be an arbitrary angle. Divide the sector $(a_1; \theta-\delta, \theta+\delta, r)$ into parts by the circumferences of the circles $c(a_1, ir\delta)$, $i = 1, 2, 3, \ldots$. The number of parts is at most $\delta^{-1}+1$, and the diameter of each part is at most $5^{\frac{1}{2}} r\delta$. Applying (25) in turn to each of these parts which contain points of $E_1$ we deduce that

$$\Lambda^s E_1.(a_1; \theta-\delta, \theta+\delta, r) < (\delta^{-1}+1) 2^{s+1} (5^{\frac{1}{2}} r\delta)^s,$$

whence

$$\frac{\Lambda^s E_1.(a_1; \theta-\delta, \theta+\delta, r)}{r^s} = O(\delta^{s-1}) \quad \text{as } \delta \to 0,$$

uniformly with respect to $r$ in the range $(0, r_1)$. It follows that (23) holds at the point $a_1$ with $E$ replaced by $E_1$. Provided that the circular density of the set $E-E_1$ is equal to zero at $a_1$, then (23) holds as it stands. Consequently (23) holds at almost all points of $E_1$, and the lemma follows from (24) because $\epsilon$ is arbitrary.

THEOREM XI. *At almost every point of any s-set whose dimension lies strictly between unity and two, the lower angular density is equal to zero in at least one angle* $(\phi, \phi+\pi)$.

*Proof.* Suppose that $E$ is any $s$-set, where $1 < s < 2$, and that $\delta < \pi$, $\lambda$, and $r_1$ are three arbitrary positive numbers. Let $E_1 = E_1(\delta, \lambda, r_1)$ denote the set of points $a$ of $E$ such that for every angle $\theta$

$$\Lambda^s E.(a; \theta, \theta+\pi-\delta, r) > \lambda r^s \quad \text{whenever } r < r_1. \tag{26}$$

Suppose, contrary to the theorem, that $\Lambda^s E_1 > 0$, and choose a closed set $F \subset E_1$ of positive measure and a positive number

$$r_2 < r_1 \tag{27}$$

such that at every point $a$ of $F$

$$\Lambda^s E.c(a, r) < 2^{s+1} r^s \quad \text{whenever } r < r_2. \tag{28}$$

Let $a_1$ be any point of $F$ at which the circular density of $E-F$ is equal to zero. Then we can, given $\epsilon > 0$, find a positive number $r_3$ such that

$$r_3 < r_2 \tag{29}$$

and

$$\Lambda^s(E-F).c(a_1, r_3) < \epsilon r_3^s. \tag{30}$$

Draw the circle $c(a_1, \frac{1}{2}r_3)$. Let $f(x, y)$ denote the distance of the point $(x, y)$ from the set $F$, and let $\rho$ denote the maximum value of this function taken over points $(x, y)$ of the circle. We shall calculate a lower bound for $\rho$. First of all we notice that

$$\rho \leqslant \tfrac{1}{2}r_3. \tag{31}$$

Let $b_1$ be an arbitrary point of $F.c(a_1, \frac{1}{2}r_3)$. Let $b_2$ be an arbitrary point of $F$ which lies inside $c(a_1, \frac{1}{2}r_3)$ but outside the circle $c(b_1, 2\rho)$. Let $b_3$ be an arbitrary point of $F$ which lies inside $c(a_1, \frac{1}{2}r_3)$ but outside both of the circles $c(b_1, 2\rho)$ and $c(b_2, 2\rho)$. We proceed in this way until the process terminates, forming a finite sequence of non-overlapping circles $c(b_i, \rho)$, $i = 1, 2, 3, \ldots, n$. It follows from (31) that these circles all lie inside $c(a_1, r_3)$, and consequently

$$\Lambda^s E.c(a_1, r_3) \geqslant \sum_{i=1}^{n} \Lambda^s E.c(b_i, \rho).$$

From (31), (29), and (27), $\rho < \frac{1}{2}r_1$, and hence we can apply (26) to each of these circles. This gives us

$$\Lambda^s E.c(a_1, r_3) > n\lambda\rho^s,$$

and using (28) and (29), we deduce that

$$2^{s+1}r_3^s > n\lambda\rho^s. \tag{32}$$

On the other hand it follows from our construction of the circles that

$$\sum_{i=1}^{n} c(b_i, 2\rho) \supset F \,.\, c(a_1, \tfrac{1}{2}r_3),$$

and hence from our definition of the number $\rho$,

$$\sum_{i=1}^{n} c(b_i, 3\rho) \supset c(a_1, \tfrac{1}{2}r_3),$$

whence $$9n\rho^2 \geqslant \tfrac{1}{4}r_3^2.$$

Eliminating $n$ by combining this inequality with (32), we have

$$9\lambda^{-1}\rho^{2-s} > 2^{-s-3}r_3^{2-s},$$

and since $s < 2$, we can write

$$\rho > K_{\lambda,s}r_3, \tag{33}$$

where $K_{\lambda,s}$ is a positive function of $\lambda$ and $s$ alone.

We have now proved the existence of a point $b$ in $c(a_1, \frac{1}{2}r_3)$ whose distance, $\rho_1$ say, from $F$ satisfies

$$K_{\lambda,s}r_3 < \rho_1 \leqslant \tfrac{1}{2}r_3. \tag{34}$$

Draw the circle $c(b, \rho_1)$, and let $p$ be the point, or one of the points, of $F$ lying on its circumference. Draw through $p$ the two chords which make an angle $\frac{1}{2}\delta$ with the tangent at $p$. Let $\rho_2$ denote the length of each of these chords. By elementary geometry,

$$\tfrac{1}{2}\delta\rho_1 < \rho_2 < 2\rho_1. \tag{35}$$

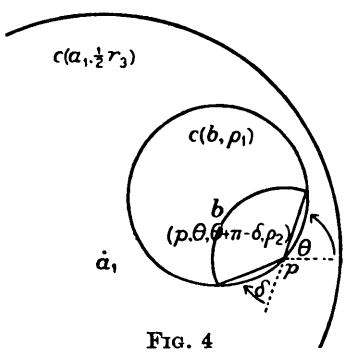


FIG. 4

(See Fig. 4.) Let $\theta$ and $\theta+\pi-\delta$ denote the directions of these chords. The sector $(p; \theta, \theta+\pi-\delta, \rho_2)$ lies inside $c(b, \rho_1)$ and this circle contains no points of $F$ in its interior. Obviously then

$$\Lambda^s F \,.\, (p; \theta, \theta+\pi-\delta, \rho_2) = 0. \tag{36}$$

Now since $b$ lies inside $c(a_1, \frac{1}{2}r_3)$, it follows from (34) that $c(b, \rho_1)$ lies inside $c(a_1, r_3)$ and therefore our sector also lies inside $c(a_1, r_3)$. This enables us to deduce from (30) that

$$\begin{aligned} \Lambda^s(E-F) \,.\, (p; \theta, \theta+\pi-\delta, \rho_2) &< \epsilon r_3^s \\ &< K_{\lambda,s}^{-s}\epsilon\rho_1^s, \quad \text{from (34)}, \\ &< K_{\delta,\lambda,s}\epsilon\rho_2^s, \quad \text{from (35)}, \end{aligned}$$

where $K_{\delta,\lambda,s}$ is a finite function of $\delta$, $\lambda$, and $s$ alone.

Combined with (36), this gives

$$\Lambda^s E \,.\, (p; \theta, \theta+\pi-\delta, \rho_2) < K_{\delta,\lambda,s}\epsilon\rho_2^s. \tag{37}$$

Now from (35), (34), (29), and (27),

$$\rho_2 < r_1. \tag{38}$$

Thus having chosen a positive number $\epsilon$ quite independently of $\delta$, $\lambda$, and $r_1$, we have found an angle $\theta$ and a positive number $\rho_2$ which satisfy (37) and (38) at a certain point $p$ of $F$ which is in turn contained in $E_1$. If $\epsilon$ is sufficiently small this is in contradiction with (26), and so our original assumption that $\Lambda^s E_1 > 0$ must be false. We deduce that at almost every point $a$ of $E$ there exists an angle $\theta = \theta(a)$ and a number $r_4 = r_4(a) < r_1$ such that

$$\Lambda^s E.(a; \theta, \theta+\pi-\delta, r_4) \leqslant \lambda r_4^s. \tag{39}$$

The numbers $\delta$, $\lambda$, and $r_1$ may be made as small as we please. For every positive integer $m$ we may set

$$\delta^{(m)} = \lambda^{(m)} = r_1^{(m)} = m^{-1},$$

and for almost every point $a$ of $E$, find an angle $\theta^{(m)} = \theta^{(m)}(a)$ and a positive number $r_4^{(m)} = r_4^{(m)}(a) < m^{-1}$ which satisfy (39). Let $\phi = \phi(a)$ denote any one of the limiting angles of the sequence $\theta^{(m)}$, $m = 1, 2, 3, \ldots$. Then provided that (23) of Lemma 29 holds for $\theta = \phi$ and $\theta = \phi+\pi$ at the point $a$ we shall have

$$\lim_{r \to 0} \Lambda^s E.(a; \phi, \phi+\pi, r) = 0.$$

This holds at almost every point of $E$ for some angle $\phi$, and so the theorem is proved.

## 11. Circular density of plane $s$-sets

11.1. In accordance with the definitions of regular and irregular points of linearly measurable sets given by Besicovitch ((2), 298), we introduce the following

DEFINITIONS. Every point of any $s$-set at which the circular density exists and is equal to unity will be called a *regular* point, and every other point an *irregular* point.

We shall call an $s$-set *regular* if almost all its points are regular, and *irregular* if almost all its points are irregular.

Besicovitch ((2), 324) has shown that almost all points of any regular linearly measurable set lie on a sequence of rectifiable curves. In particular, a rectifiable curve affords a very simple example of such a set. It is well known that any plane measurable set is a regular 2-set in the sense of our definition. Finally, and rather trivially, any 0-set consists of a finite number of points and every point is regular.

Thus it is easy to find regular $s$-sets for integral values of $s$. In this section we shall prove that regular $s$-sets for non-integral values of $s$ do not exist. The problem was suggested to me by Professor A. S. Besicovitch.

11.2. The following lemma is already known, but has not been published.

LEMMA 30. *At almost every point of any s-set, where* $0 < s < 1$, *the circular density fails to exist.*

*Proof.* Suppose that there exists an $s$-set $E$, where $0 < s < 1$, for which our lemma is false. Then by Lemma 3 we can find a set of positive measure $E_1 \subset E$ and a positive number $r_1$ such that at every point $a$ of $E_1$, the circular density of $E$ exists and

$$\Lambda^s E.c(a,\rho) > \tfrac{1}{2}\rho^s \quad \text{whenever } \rho < r_1. \tag{1}$$

Let $a_1$ be any point of $E_1$ at which the circular density of $E-E_1$ is equal to zero. Then the circular densities of both $E$ and $E_1$ at $a$ exist and are equal to each other. Let their common value be $2^{-s}\mu = 2^{-s}\mu(a_1)$. We can, given $\epsilon > 0$, find a positive number $r_2 < r_1$ such that whenever $\rho < 2r_2$,

$$(\mu-\epsilon)\rho^s < \Lambda^s E_1.c(a_1,\rho) \leqslant \Lambda^s E.c(a_1,\rho) < (\mu+\epsilon)\rho^s. \tag{2}$$

Suppose that $\alpha < 1$ is an arbitrary positive number. From (2)

$$\Lambda^s E.c(a_1,r_2) > (\mu-\epsilon)r_2^s$$

and

$$\Lambda^s E.c(a_1,(1+\alpha)r_2) < (\mu+\epsilon)(1+\alpha)^s r_2^s,$$

and consequently

$$\Lambda^s E.[c(a_1,(1+\alpha)r_2)-c(a_1,r_2)] < (\mu+\epsilon)(1+\alpha)^s r_2^s-(\mu-\epsilon)r_2^s. \tag{3}$$

By a similar argument, we also deduce from (2) that

$$\Lambda^s E_1.[c(a_1,(1+\tfrac{2}{3}\alpha)r_2)-c(a_1,(1+\tfrac{1}{3}\alpha)r_2)] > (\mu-\epsilon)(1+\tfrac{2}{3}\alpha)^s r_2^s-(\mu+\epsilon)(1+\tfrac{1}{3}\alpha)^s r_2^s,$$

which is positive if we have chosen any sufficiently small value of $\epsilon$. In this case there exists a point $a_2$ of $E_1$ lying in the annulus

$$c(a_1,(1+\tfrac{2}{3}\alpha)r_2)-c(a_1,(1+\tfrac{1}{3}\alpha)r_2),$$

and so the circle $c(a_2,\frac{1}{3}\alpha r_2)$ lies inside the annulus

$$c(a_1,(1+\alpha)r_2)-c(a_1,r_2).$$

Using (1), it follows that

$$\Lambda^s E.[c(a_1,(1+\alpha)r_2)-c(a_1,r_2)] > \tfrac{1}{2}(\tfrac{1}{3}\alpha r_2)^s,$$

and together with (3) this gives

$$(\mu+\epsilon)(1+\alpha)^s-(\mu-\epsilon) > \tfrac{1}{6}\alpha^s.$$

Since $\epsilon$ can be made as small as we please, we must have

$$\mu(1+\alpha)^s-\mu \geqslant \tfrac{1}{6}\alpha^s.$$

On the other hand,

$$(1+\alpha)^s = 1+O(\alpha) \quad \text{as } \alpha \to 0,$$

and we have arrived at a contradiction. The lemma now follows.

11.3. It remains to consider the case of $s$-sets whose dimensions lie strictly between unity and two. We shall prove that at almost all points of such a set, the lower circular density is less than unity. The reader may notice that by a more careful analysis it would be possible, for any given value of $s$, to assign to this density an upper bound. This bound, however, would be a number very near unity, especially when $s$ is near unity, and in any case we cannot expect it to represent the best possible result.

First of all we introduce the following

DEFINITION. An $s$-set $E$ will be said to possess a *weak tangent* at the point $a$ if: (i) the lower circular density at $a$ is positive, and (ii) there exists an angle $\theta$ such that for any pair of positive numbers $\delta$ and $\epsilon$ there exist arbitrarily small values of $r$ such that

$$\Lambda^s E.(a;\theta+\delta,\theta+\pi-\delta,r)+\Lambda^s E.(a;\theta+\pi+\delta,\theta-\delta,r) < \epsilon r^s. \qquad (4)$$

The weak tangent at $a$ is, of course, the line through $a$ which lies in the direction $\theta$.

Notice that if we altered the definition by saying that (4) must be satisfied by *all sufficiently small* values of $r$, we would have defined a tangent equivalent to that defined by Besicovitch ((**2**), 298).

Notice also that a set can possess any number of weak tangents at a single point.

Now from Lemma 29 we deduce immediately

LEMMA 31. *At almost every point of any $s$-set, where $s > 1$, a weak tangent fails to exist.*

On the other hand we shall prove

LEMMA 32. *At almost every regular point of any $s$-set $E$, where $1 < s < 2$, a weak tangent exists.*

*Proof.* We shall show, in fact, that at every regular point of $E$ at which the convex density is equal to unity, a weak tangent exists in every direction $\theta$ for which the lower angular density is zero in the angle $(\theta+\pi, \theta+2\pi)$, and the lemma will follow by using Lemma 1 and Theorem XI.

Suppose then that $a$ is any such point, and that $\theta$ is a suitable direction. Then we can, given any positive number $\eta$, find an arbitrarily small positive number $r_1$ such that

$$\Lambda^s E.c(a,r) > 2^s(1-\eta)r^s \quad \text{whenever } r \leqslant r_1, \qquad (5)$$

$$\Lambda^s EU < (1+\eta)(dU)^s \quad \text{whenever } a \in U \text{ and } dU < r_1, \qquad (6)$$

and $$\Lambda^s E.(a, \theta+\pi, \theta+2\pi, r_1) < 2^s \eta r_1^s. \tag{7}$$

Draw the semicircle $(a, \theta, \theta+\pi, r_1)$, and let the end points of the diameter be denoted by $p_1$ and $p_1'$ (as in Fig. 5). Suppose that $\delta$ is an arbitrary positive number less than $\frac{1}{2}\pi$. Draw the rays $(a; \theta+\delta)$ and $(a; \theta+\pi-\delta)$ and let them meet the circumference in $q_1$ and $q_1'$ respectively. Now by a simple continuity argument, there exists a point lying on $(a; \theta)$ whose distance from $p_1'$ is equal to the sum of its distances from $p_1$ and $q_1'$. Denote this point by $p_2$. With centre at $p_2$, draw an arc through $q_1'$ to meet $(a; \theta+\pi)$ in $p_2'$. Then $p_1p_2 = p_1'p_2'$ and so by symmetry it follows that we can, with

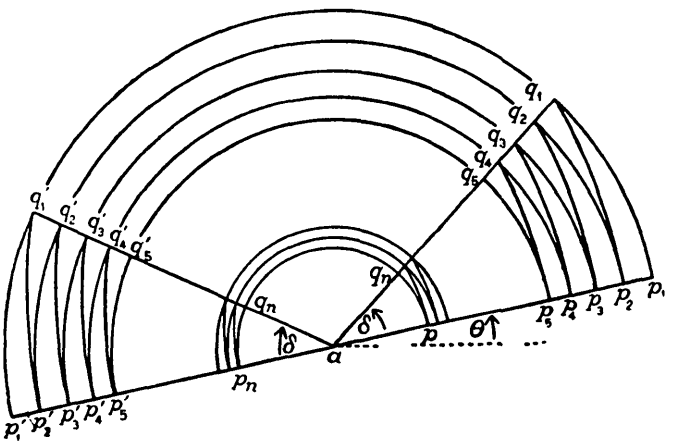



FIG. 5

centre at $p_2'$, draw an arc through $p_2$ and $q_1$. Denote by $U_1$ the convex area cut off from our semicircle by the two arcs which we have drawn. Then $dU_1 = 2ap_2 = 2r_2$, say.

Now draw the semicircle $(a; \theta, \theta+\pi, r_2)$. Then $p_2$ and $p_2'$ are the end points of its diameter. Let the rays $(a; \theta+\delta)$ and $(a; \theta+\pi-\delta)$ meet its circumference in $q_2$ and $q_2'$ respectively. Upon our new semicircle perform exactly the same operation as that which was performed upon $(a; \theta, \theta+\pi, r_1)$, using the points $p_2, p_2', q_2$, and $q_2'$. In this way, form the convex area $U_2$, $dU_2 = 2r_3$, and the semicircle $(a; \theta, \theta+\pi, r_3)$.

Let $n$ be an arbitrary positive integer, and repeat the process $n-1$ times, forming a sequence of convex areas $U_i$, $i = 1, 2, 3, \ldots, n-1$, and a sequence of semicircles $(a; \theta, \theta+\pi, r_i)$, $i = 1, 2, 3, \ldots, n$, such that

$$dU_i = 2r_{i+1} \tag{8}$$

and

$$(a; \theta+\delta, \theta+\pi-\delta, r_i) - (a; \theta+\delta, \theta+\pi-\delta, r_{i+1}) \subset U_i - (a; \theta, \theta+\pi, r_{i+1}),$$
$$i = 1, 2, 3, \ldots, n-1. \tag{9}$$

From (9)

$$\begin{aligned}\Lambda^s E.(a;\theta+\delta,\theta+\pi-\delta,r_i)-\Lambda^s E.(a;\theta+\delta,\theta+\pi-\delta,r_{i+1})&\\ \leqslant \Lambda^s EU_i-\Lambda^s E.(a;\theta,\theta+\pi,r_{i+1})&\\ = \Lambda^s EU_i-\Lambda^s E.c(a,r_{i+1})+\Lambda^s E.(a;\theta+\pi,\theta+2\pi,r_{i+1})&\\ < (1+\eta)(dU_i)^s-2^s(1-\eta)r_{i+1}^s+2^s\eta r_1^s, \quad &\text{from (5), (6), and (7),}\\ = 2^{s+1}\eta r_{i+1}^s+2^s\eta r_1^s, \quad &\text{from (8),}\\ < 2^{s+2}\eta r_1^s.&\end{aligned}$$

Summing each side over the integers $i = 1, 2, 3,\ldots, n-1$, and then transposing the second term on the left-hand side, we have

$$\begin{aligned}\Lambda^s E.(a;\theta+\delta,\theta+\pi-\delta,r_1) &< 2^{s+2}\eta(n-1)r_1^s+\Lambda^s E.(a;\theta+\delta,\theta+\pi-\delta,r_n)\\ &< 2^{s+2}\eta n r_1^s+2^s(1+\eta)r_n^s, \quad \text{from (6).} \qquad (10)\end{aligned}$$

Now $\eta$, $\delta$, and $n$ were chosen quite independently. By the nature of our construction, the ratio $r_n/r_1$ is a function of $\delta$ and $n$ alone, and for every value of $\delta$ it tends to zero as $n$ tends to infinity. Consequently, supposing $\epsilon$ to be an arbitrary positive number, we may assume that $\eta$ and $n$ have been chosen in such a way as to ensure that the right-hand side of (10) is less than $\epsilon r_1^s$. The number $\delta$ may still be regarded as arbitrary. Together with (7), this proves the lemma.

Lemmas 30, 31, and 32 give us

THEOREM XII. *Any s-set whose dimension is not an integer is irregular.*

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## REFERENCES

1. A. S. BESICOVITCH, 'On the fundamental geometrical properties of linearly measurable plane sets of points, I', *Math. Ann.* 98 (1927), 422–64.
2. —— 'On the fundamental geometrical properties of linearly measurable plane sets of points, II', ibid. 115 (1938), 296–329.
3. —— 'On the fundamental geometrical properties of linearly measurable plane sets of points, III', ibid. 116 (1939), 349–57.
4. —— 'Linear sets of fractional dimensions, I', ibid. 101 (1929), 161–93.
5. —— and P. A. P. MORAN, 'The measure of product and cylinder sets', *J. London Math. Soc.* 20 (1945), 110–20.
6. R. O. DAVIES, 'Subsets of finite measure in analytic sets', *Kon. Ned. Akad. Wet., Indag. Math.* 14 (1952), 488–9.
7. —— 'On accessibility of plane sets and approximate differentiation of functions of two real variables', *Proc. Cambridge Phil. Soc.* 48 (1952), 215–32.
8. D. R. DICKINSON, 'Study of extreme cases with respect to the densities of irregular linearly measurable plane sets of points', *Math. Ann.* 116 (1939), 358–73.

9. H. Federer, 'The $(\phi, k)$ rectifiable subsets of $n$-space', *Trans. Amer. Math. Soc.* 62 (1947), 114–89.
10. J. Gillis, 'Note on the projection of irregular linearly measurable plane sets of points', *Fund. Math.* 26 (1936), 229–33.
11. —— 'On the projection of irregular linearly measurable sets of points', *Proc. Cambridge Phil. Soc.* 30 (1934), 47–54.
12. —— 'A theorem on irregular linearly measurable sets of points', *J. London Math. Soc.* 10 (1935), 234–40.
13. P. A. P. Moran, 'On plane sets of fractional dimensions', *Proc. London Math. Soc.* 51 (1949), 415–23.
14. S. Saks, *Theory of the Integral* (Warsaw, 1937).

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